

Conceitos Básicos de Tromboelastometria



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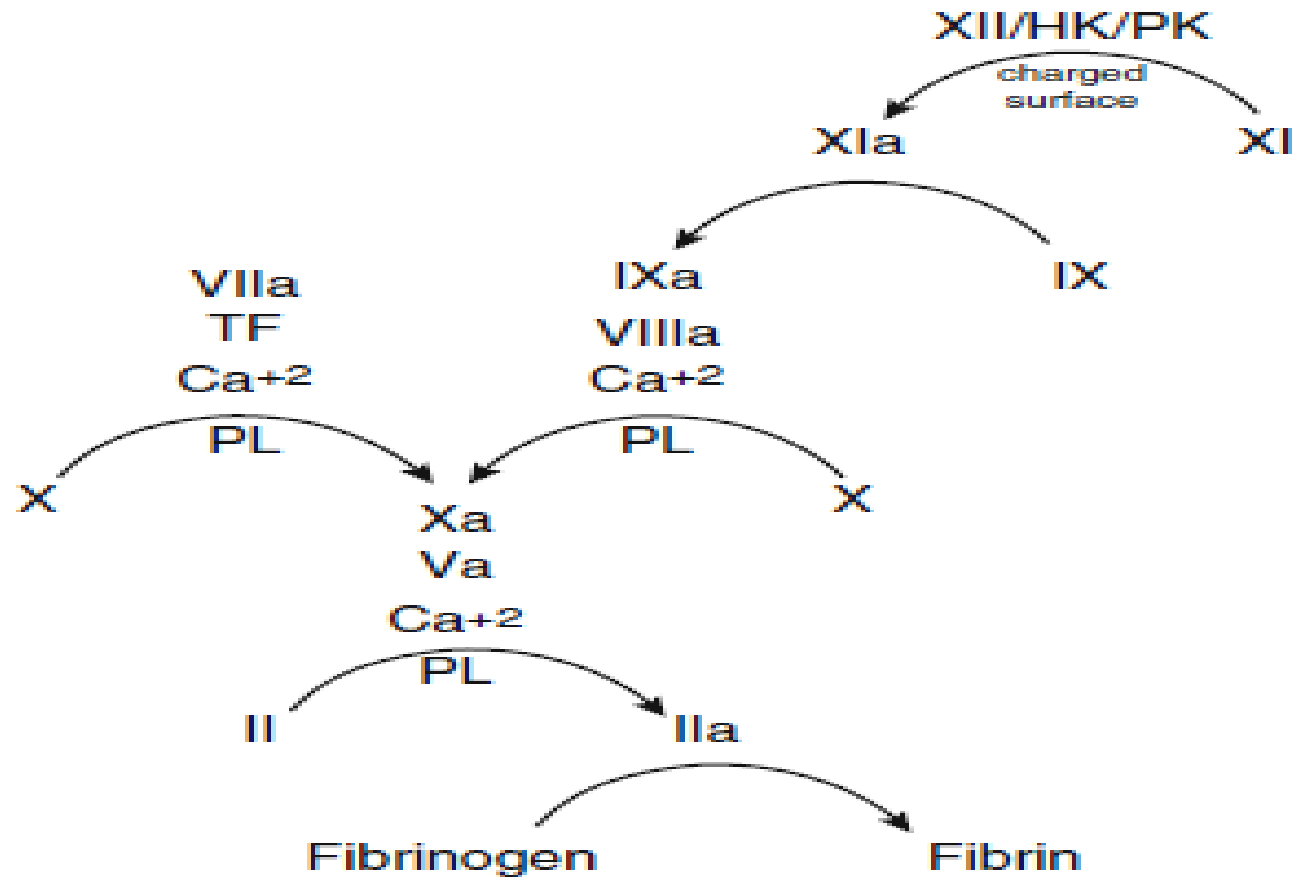
Doutor em Ciências Médicas - HCFMUSP



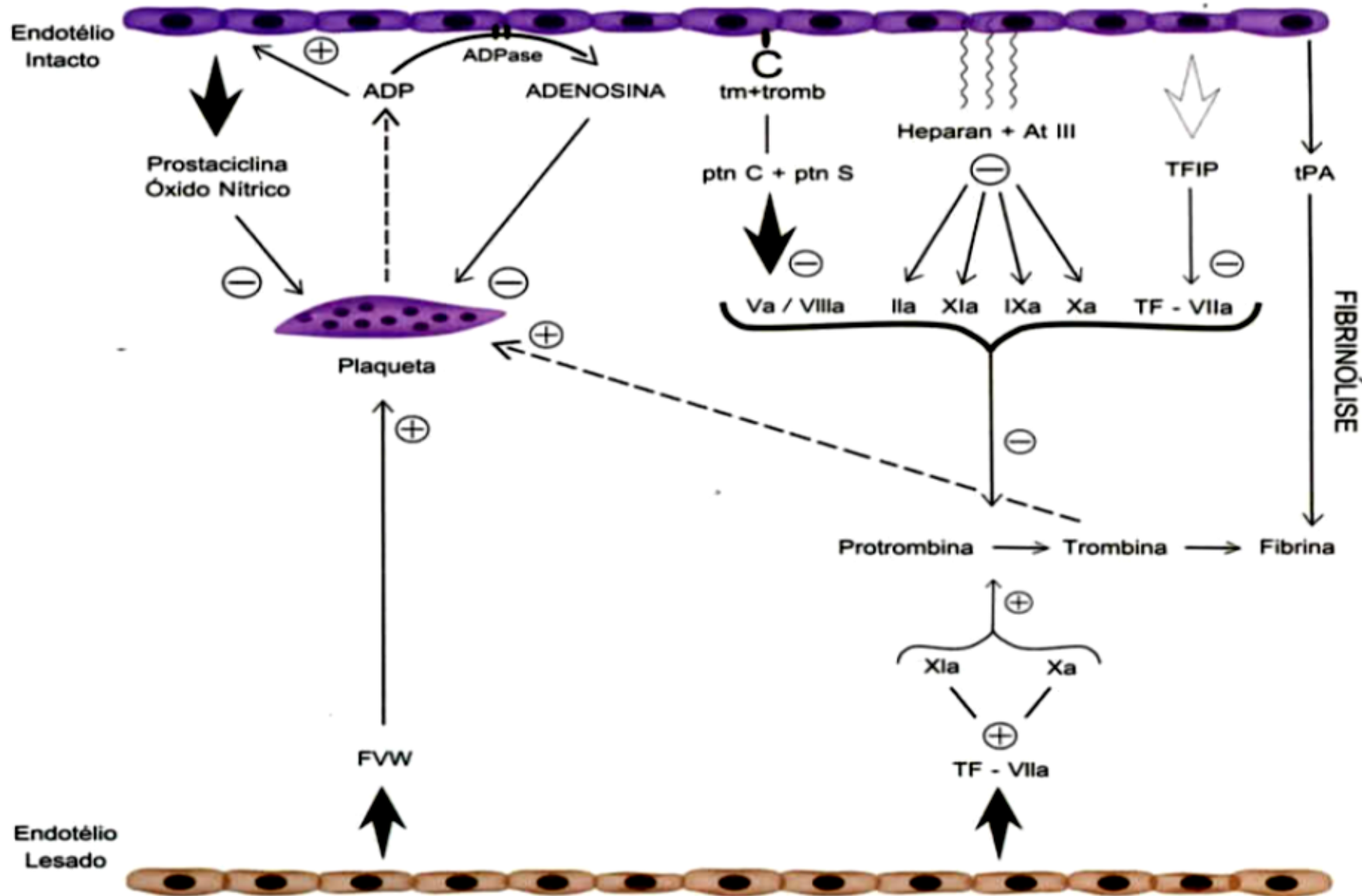
0 clásico

Extrinsic pathway PT

Intrinsic pathway aPTT



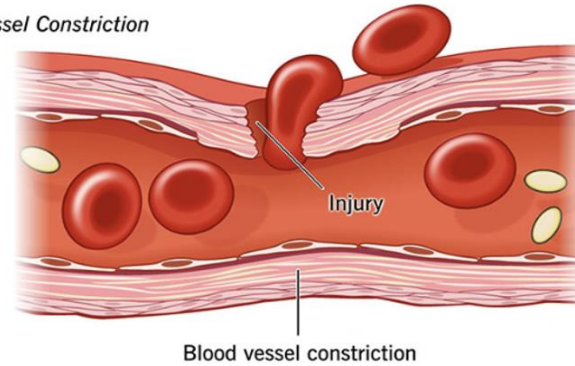
O Atual ...



Simplificando....

Lesão
+
Exposição do fator tecidual

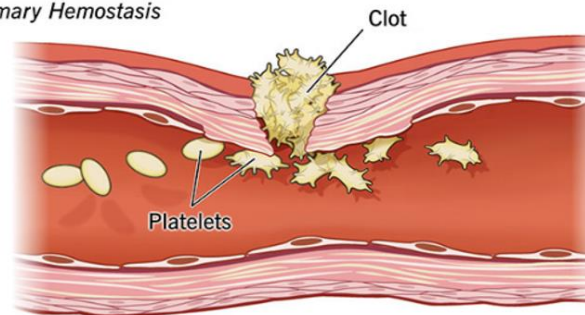
A) Vessel Constriction



Fatores de coagulação

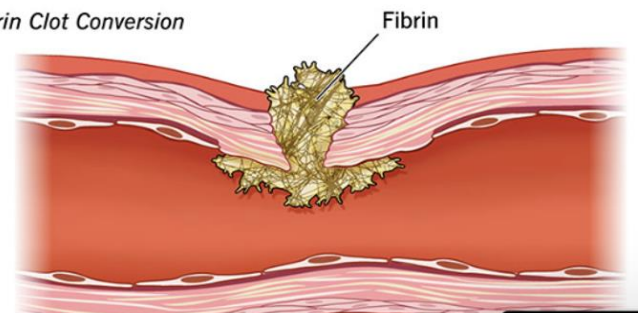
Plug Plaquetário

B) Primary Hemostasis



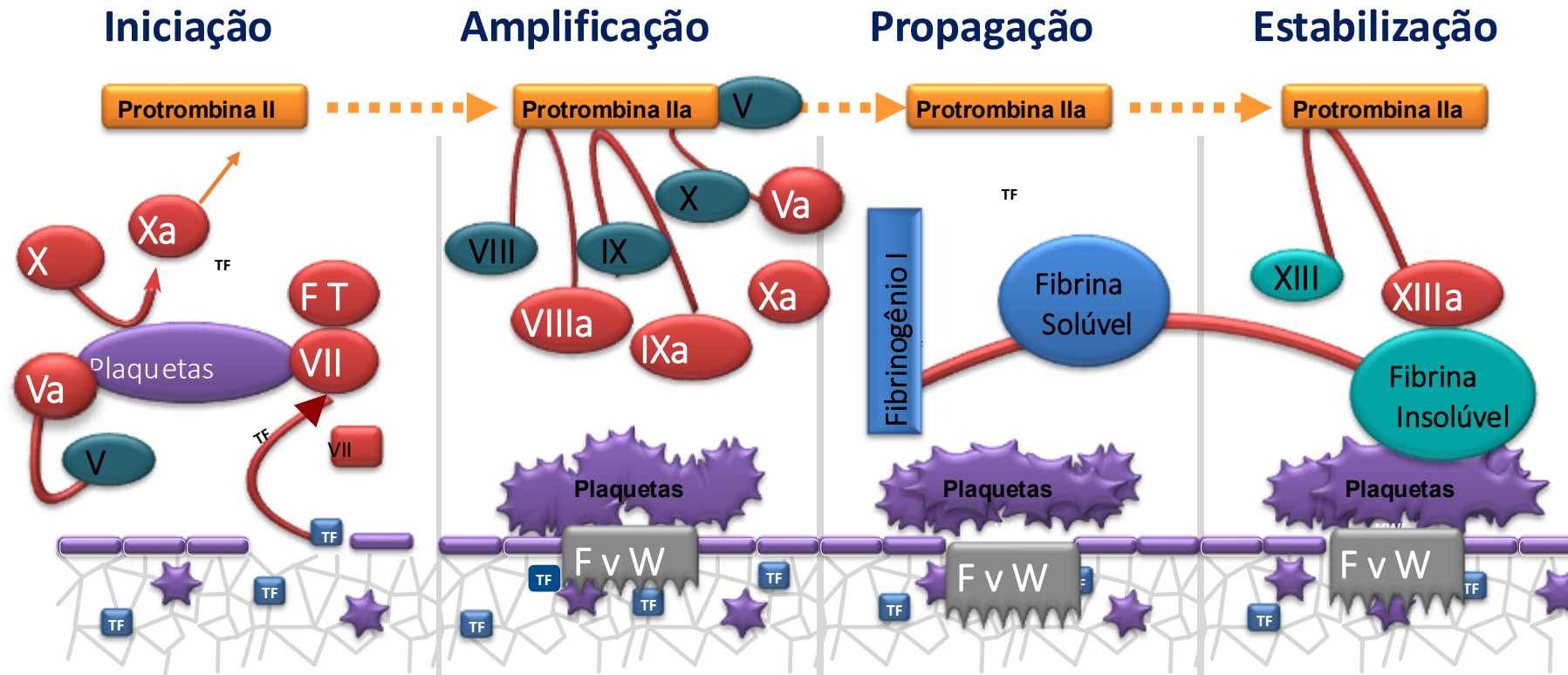
Fibrinogênio + Plaquetas

C) Fibrin Clot Conversion



Tromboelastografia

TP e TTPA



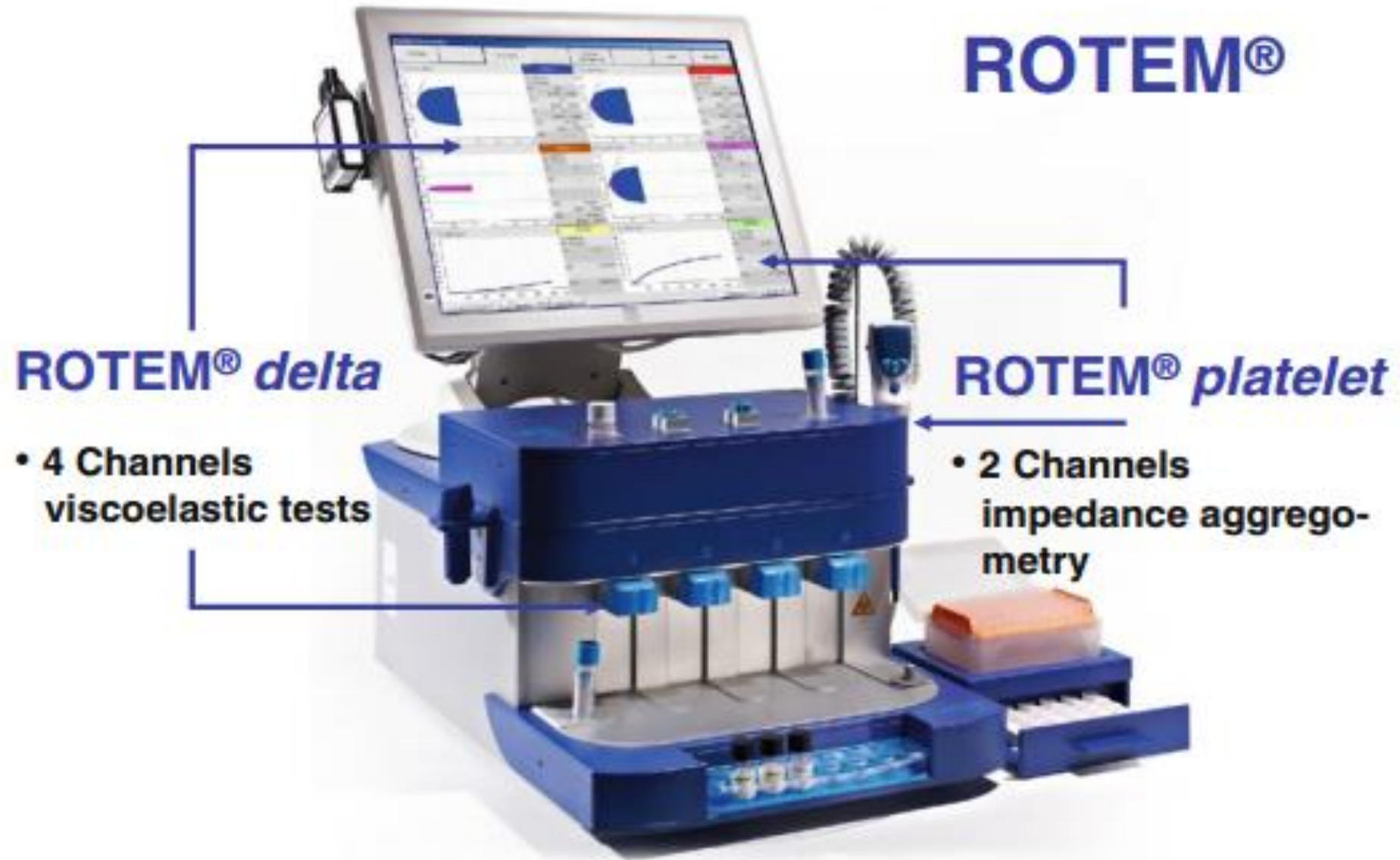
Intraoperative Hypercoagulability During Liver Transplantation as Demonstrated by Thromboelastography

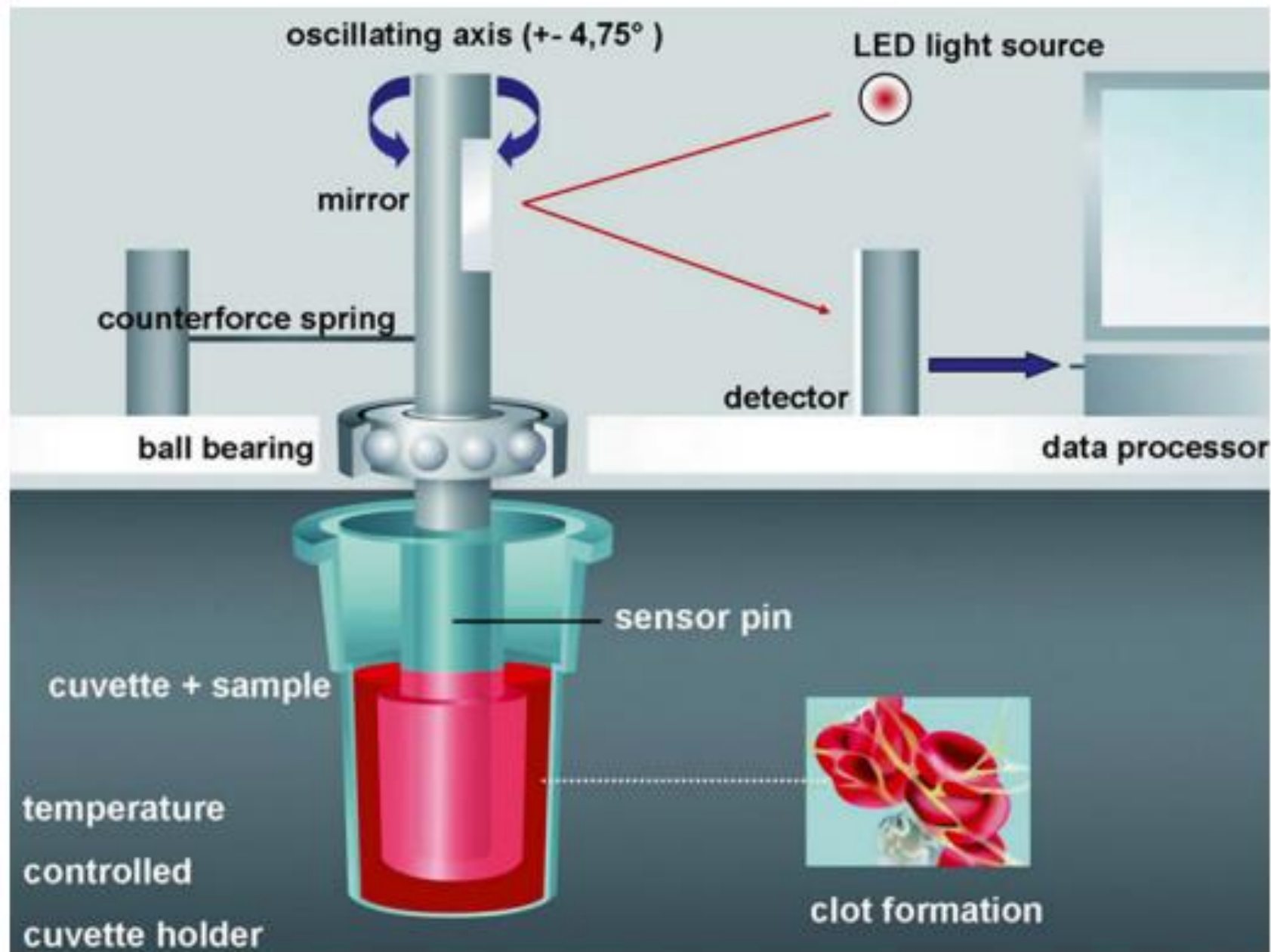
Dominik Krzanicki, Anita Sugavanam, and Susan Mallett

Department of Anaesthesia, Royal Free Hospital, London, United Kingdom

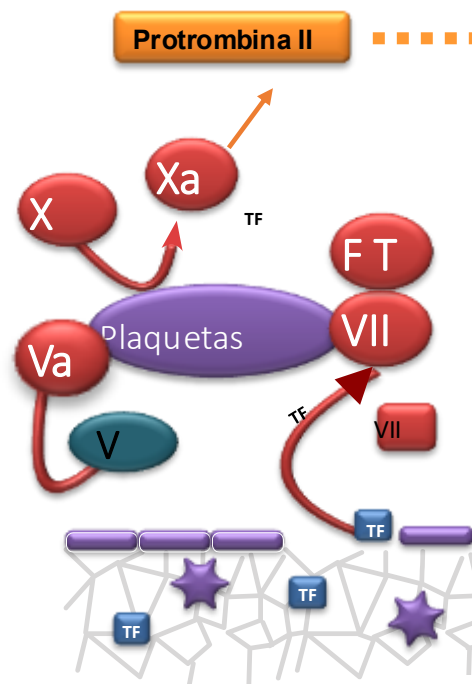
of standard plasma coagulation tests, as underlined in the Baveno VI guidelines^[21], comes from the fact they are performed without thrombomodulin addition, which is the main protein C activator, and they only detect the first 5% of whole thrombin formation. The



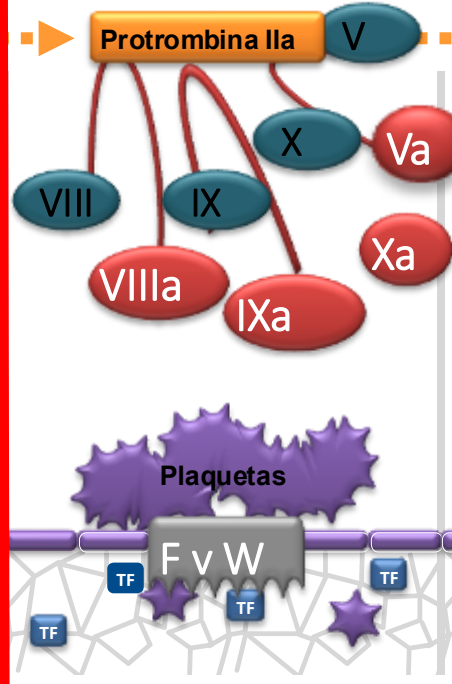




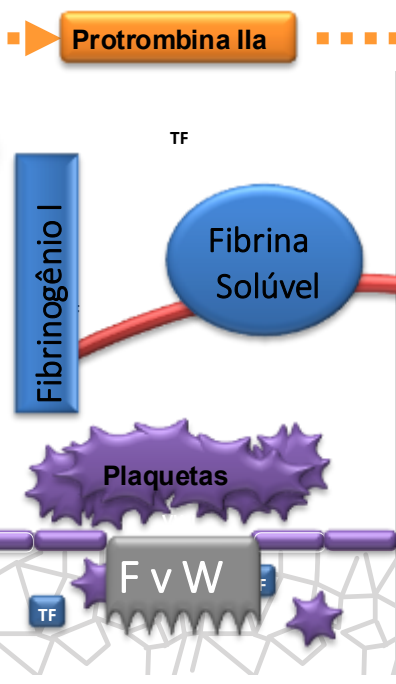
Iniciação



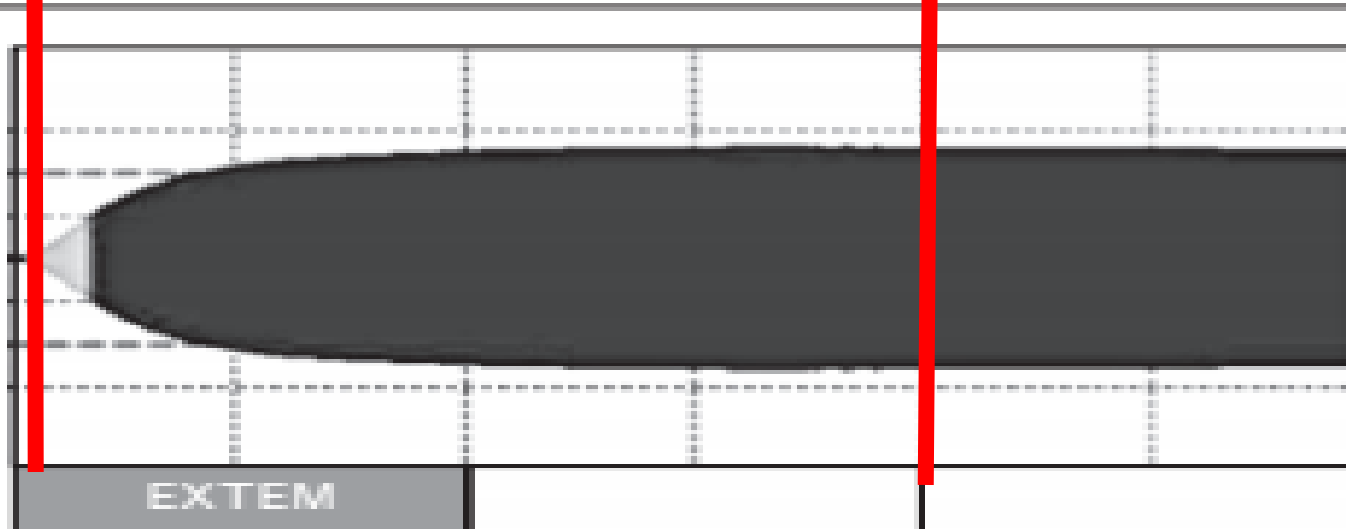
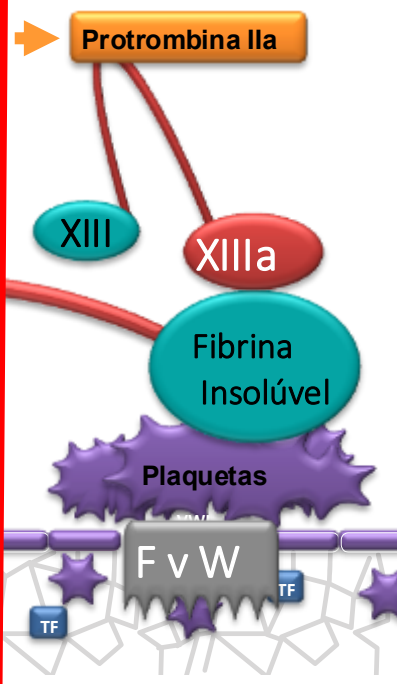
Amplificação



Propagação



Estabilização



Nomenclatura ...

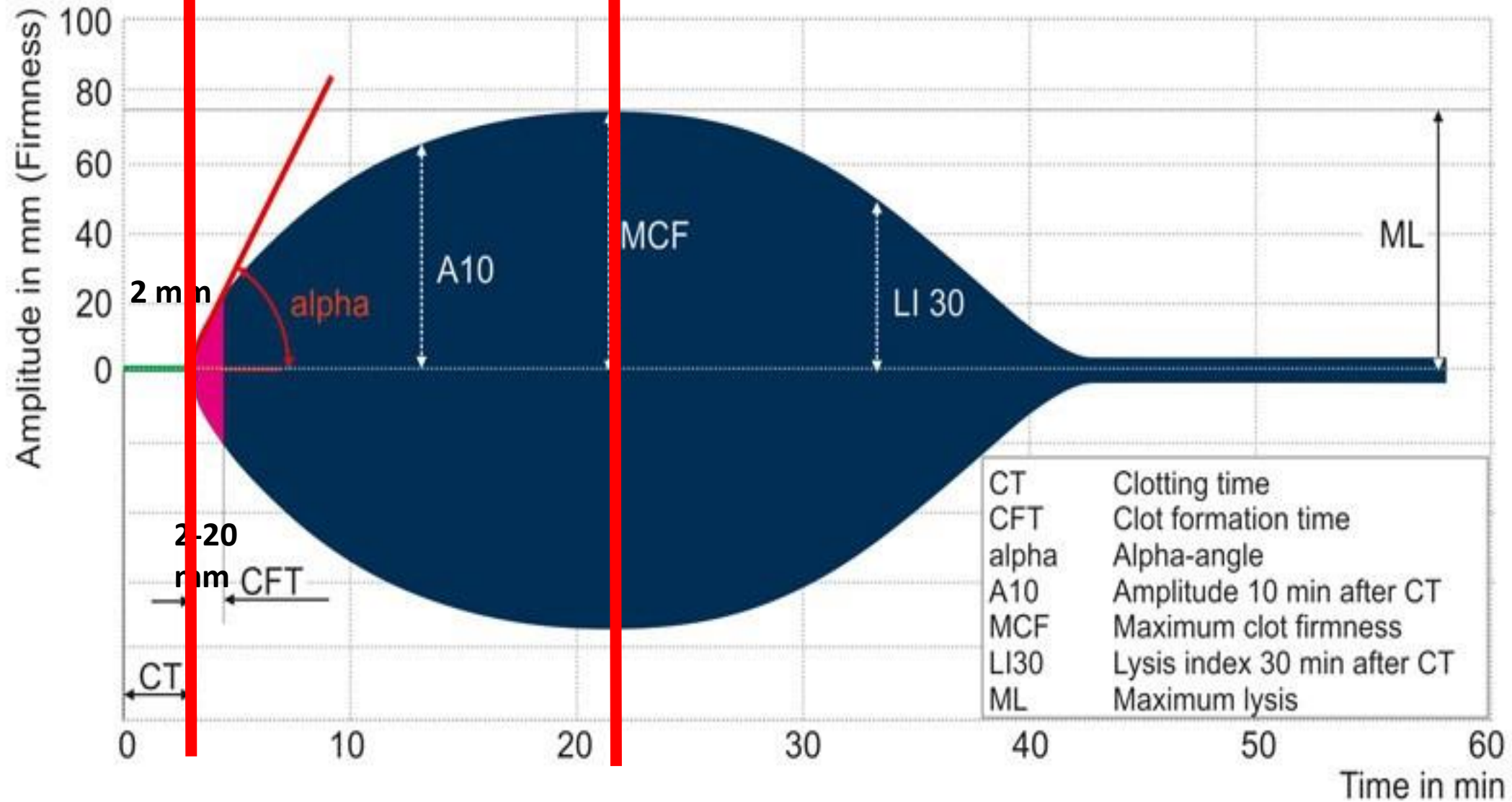


Table 2. Reference Ranges of Different Thromboelastographic Parameters

	Test (activator)	R/CT (s)	K/CFT (s)	Angle α (°)	MA/MCF (mm)	ML (% of MA)
TEG	rapidTEG (tissue factor)	78-110	30-120	68-82	54-72	<15
	kaoTEG (kaolin)	180-480	60-180	55-78	51-69	<15
ROTEM	EXTEM (tissue factor)	35-80	35-160	63-81	53-72	<15
	INTEM (ellagic acid)	100-240	35-110	71-82	53-72	<15
	FIBTEM (tissue factor + cytochalasin-D)				9-25	<15

NOTE. Normal values are according to the manufacturer of ROTEM and TEG for citrated and recalcified blood samples.





Rotational thromboelastometry-guided trauma resuscitation

Damian D. Keene, Giles R. Nordmann, and Tom Woolley

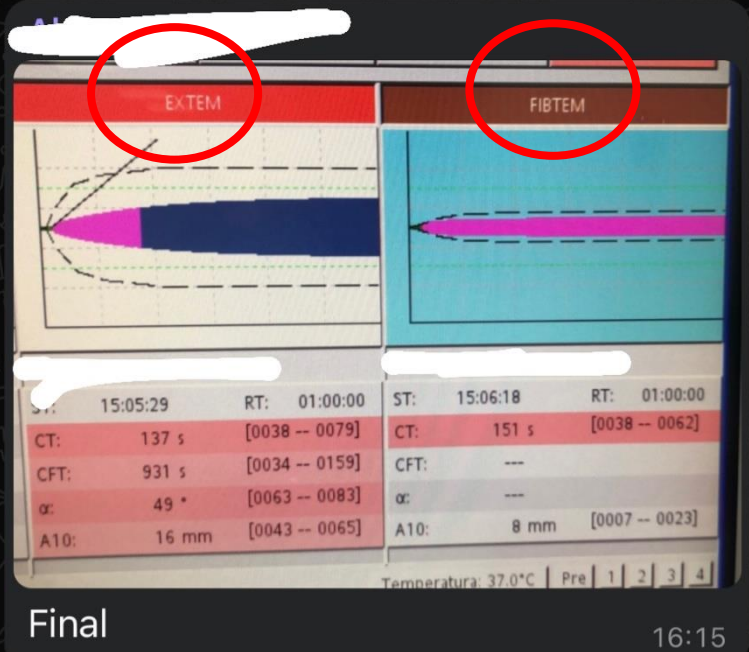
- **EXTEM** -> Mimetiza a via extrínseca -> Adição de cálcio e fator tecidual
- **FIBTEM** -> Adiciona inibidor de plaqueta (citocalasina D)
- **INTEM** -> Mimetiza a via intrínseca -> Adiciona ácido elárgico
- **APTEM** -> Adição de aprotinina -> inibe plasmina (fibrinolise?)
- **HEPTEM** -> Adiciona heparinase para avaliar efeito da heparina

Pós Operatório de Transplante Hepático

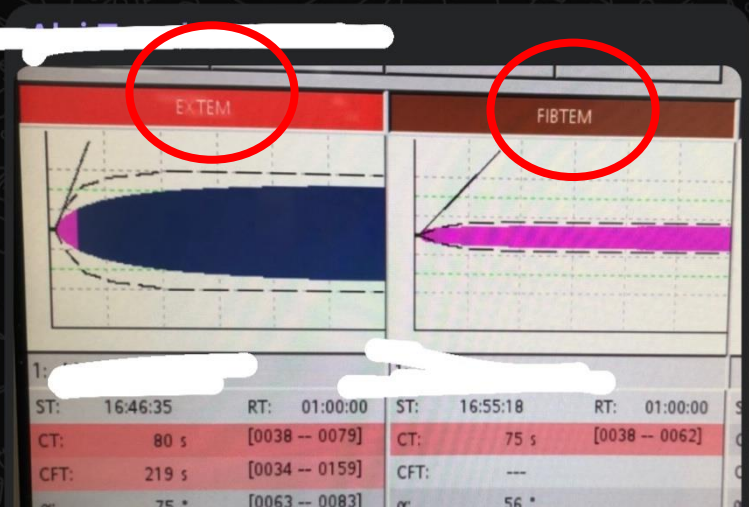
Coagulopatia x Cirúrgico ??

Parcial

15:31



Mensagem não lida: 1



Fatores de Coagulação
ou
Plaquetas

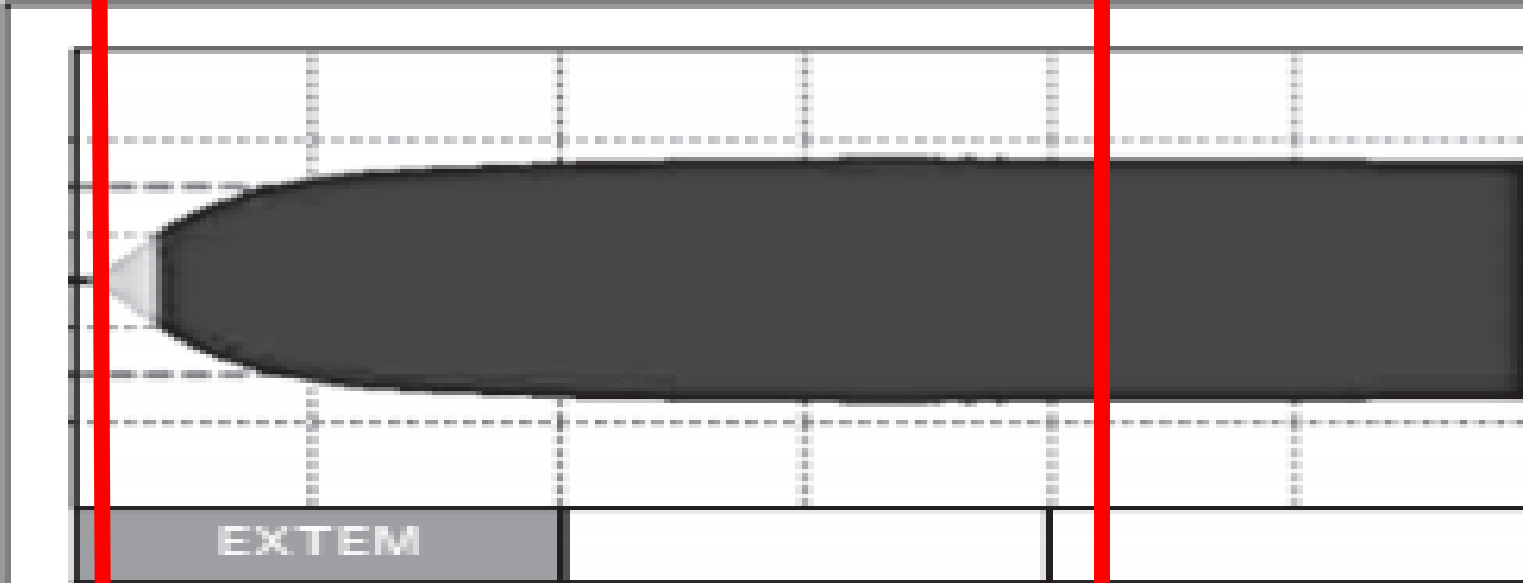
Fibrinogênio
ou
Plaquetas

Fibrinólise

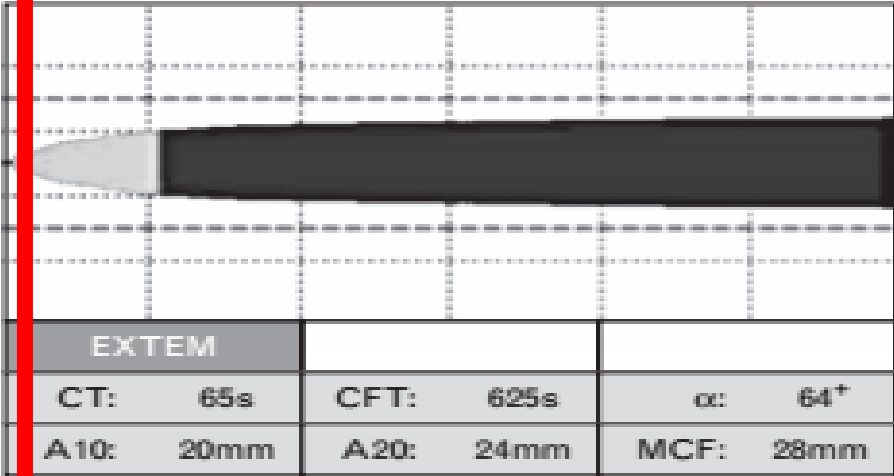
CT

CFT
Alfa
MCF

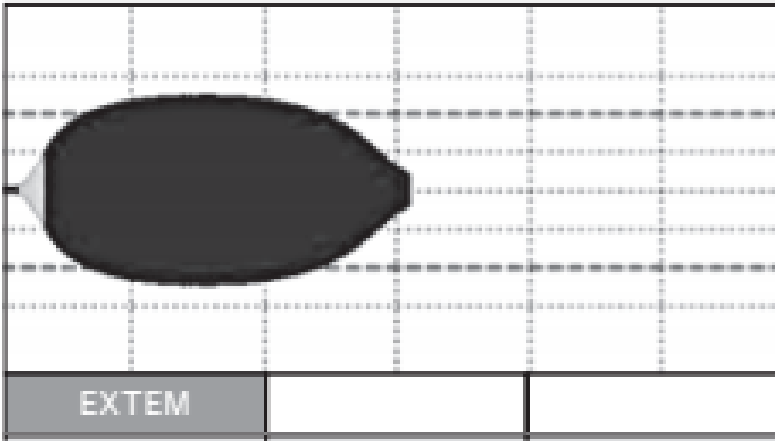
Major Lyse



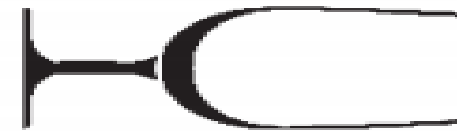
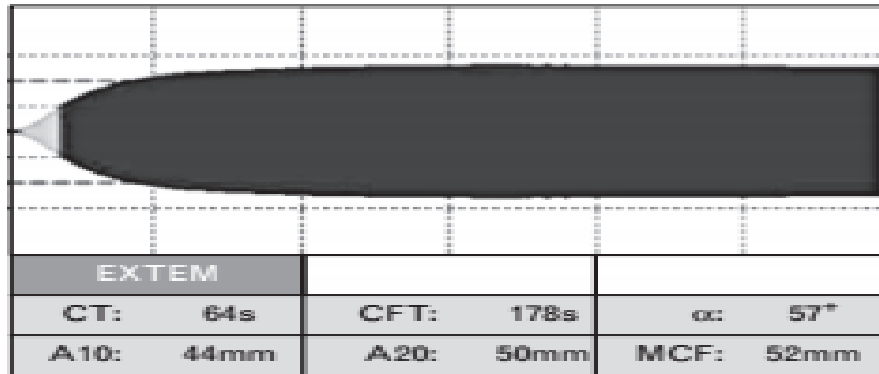
Fatores de Coaulação
ou
Plaquetas



Fibrinogênio
ou
Plaquetas



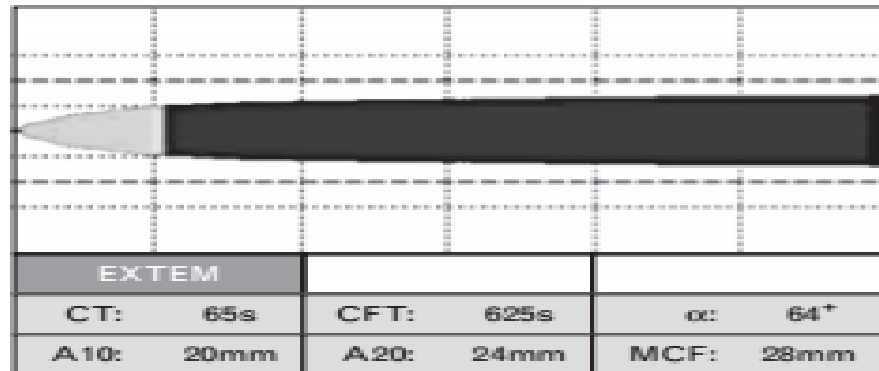
Fibrinólise



Red wine glass

Normal trace

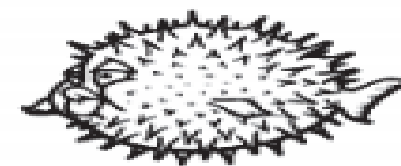
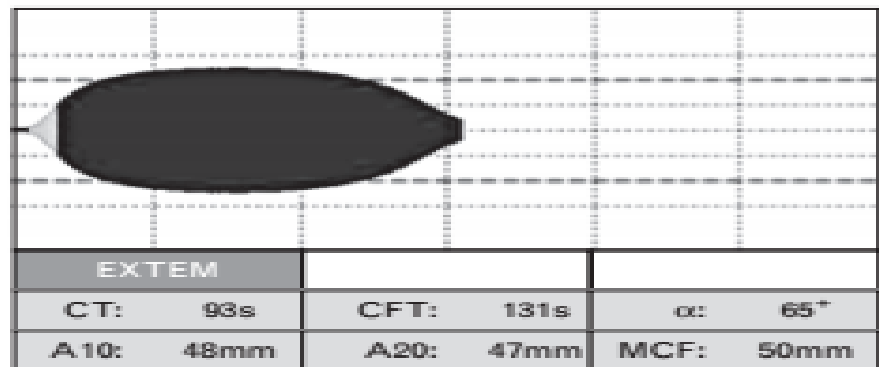
- Short CT (stem)
- Wide MCF (body)



Champagne glass

Coagulopathic trace

- Long CT (stem)
- Narrow MCF (body)

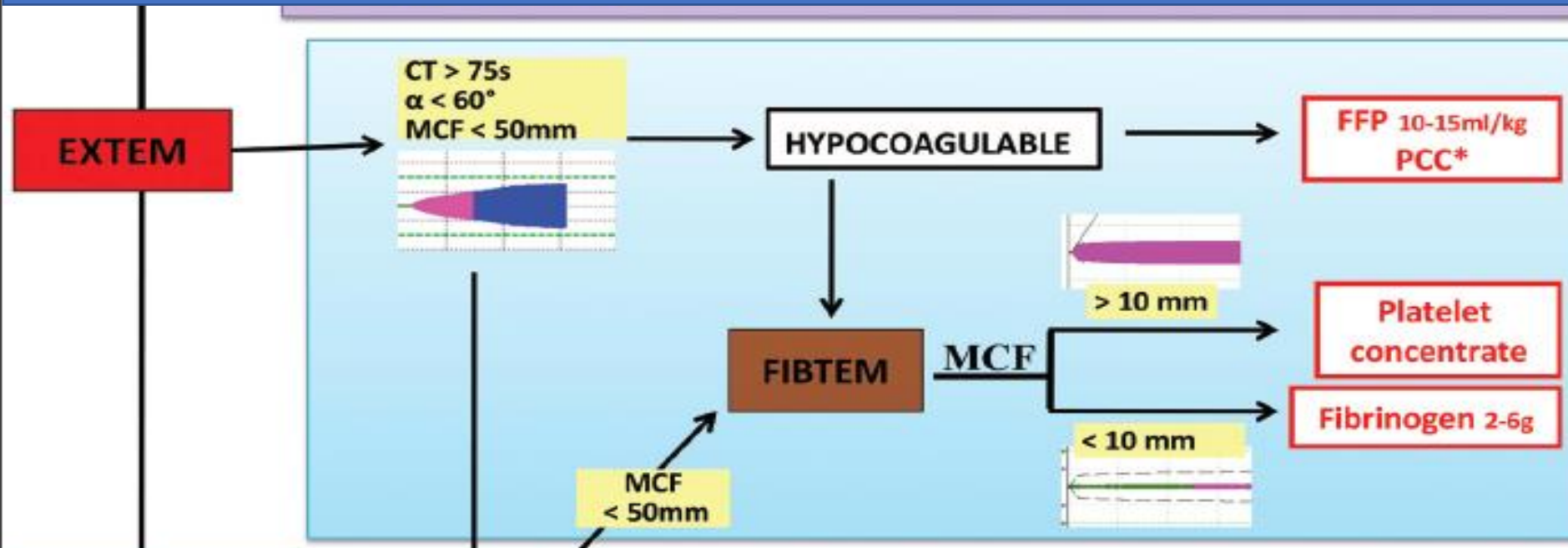


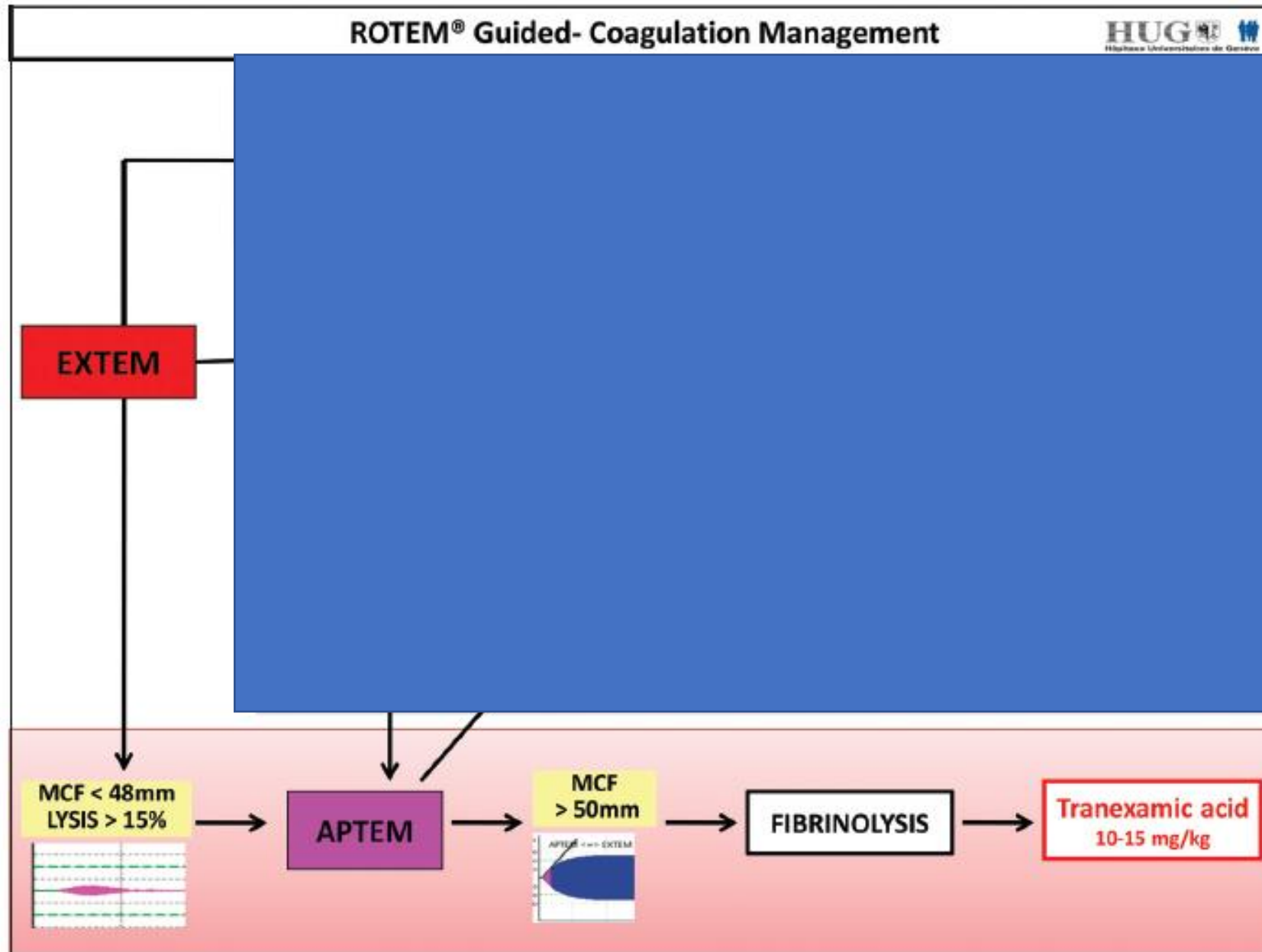
Pufferfish

Hyperfibrinolysis trace

- Wide MCF initially with narrowing

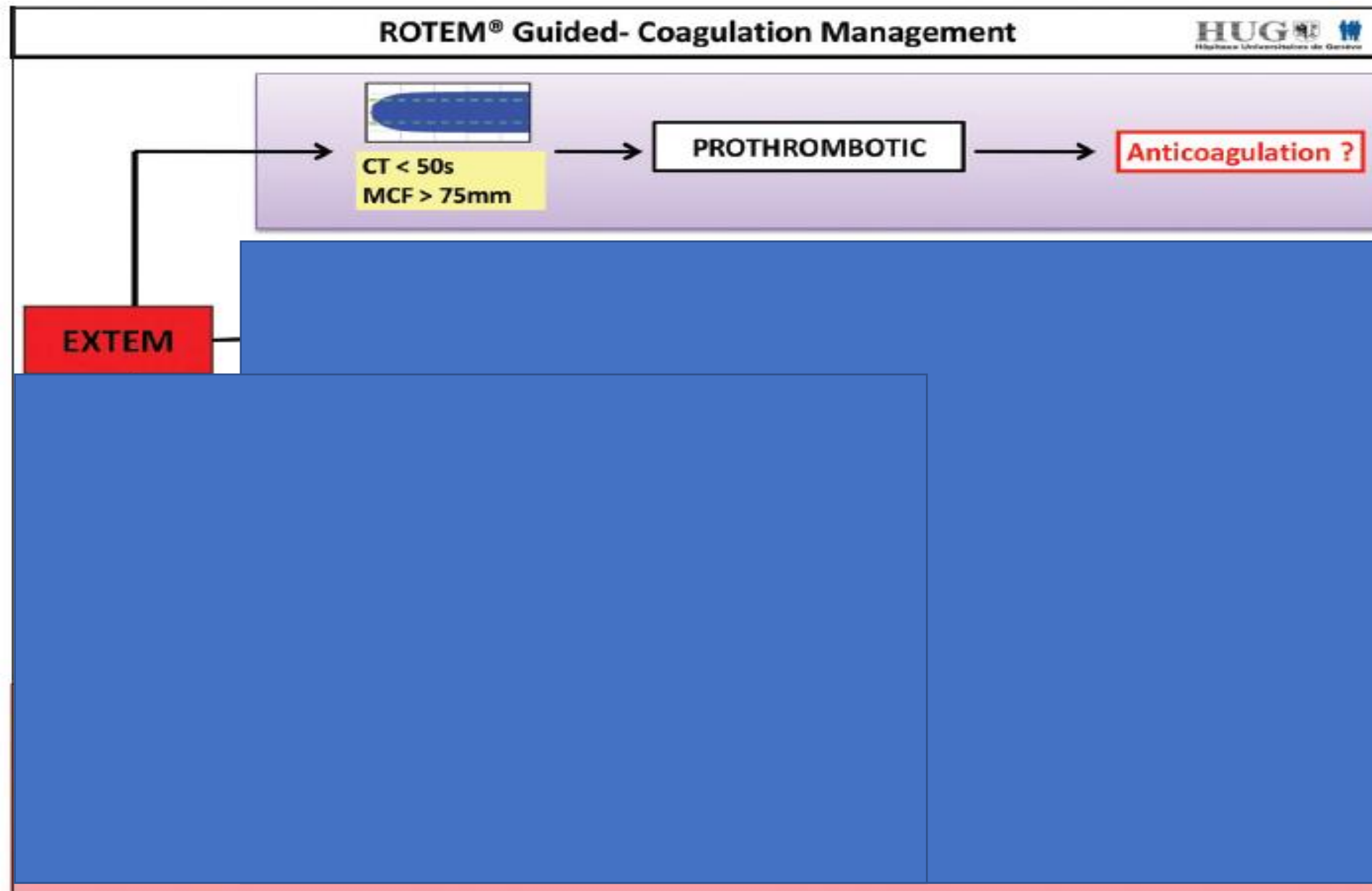
coagulopathy, a FIBTEM amplitude after 10 minutes (A10) of less than 5 mm was reported to be a good predictor of low plasma fibrinogen (<100 mg/dL) with a sensitivity of 91% and a specificity of 85% [45]. It is suggested that hetastarches and





Sartorius, et al 2014

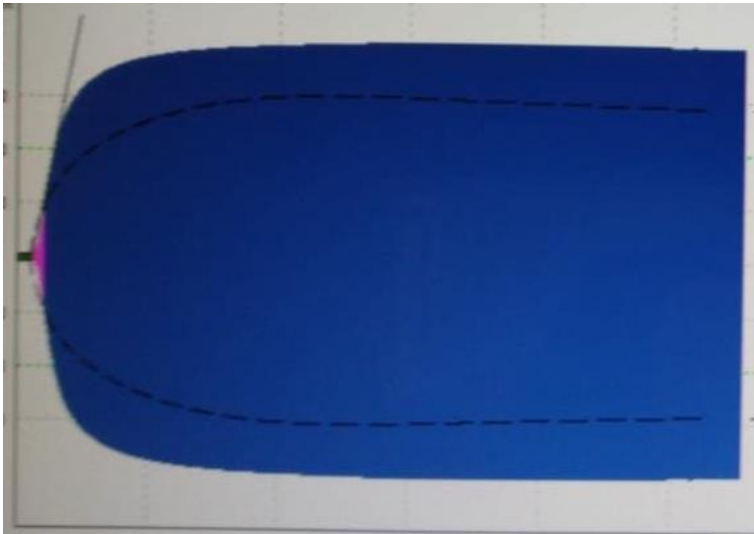




Intraoperative Hypercoagulability During Liver Transplantation as Demonstrated by Thromboelastography

Dominik Krzanicki, Anita Sugavanam, and Susan Mallett
Department of Anaesthesia, Royal Free Hospital, London, United Kingdom

**Trombo
Hipercoagulante**



Among the 117 primary transplants (ie, with the exclusion of the 7 regraft procedures), there were 6 cases (5%) of HAT. Four of the 6 required a regraft procedure as a result of HAT. Three of the 6 cases had high G traces during their initial transplants [$P=0.25$ (χ^2)], and 4 of the 6 cases demonstrated shortened R times [$P=0.79$ (χ^2)].

20% na Pop. Geral x 66% nos casos de HAT



NIH Public Access

Author Manuscript

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Int J Stroke. 2015 February ; 10(2): 194–201. doi:10.1111/j.1747-4949.2012.00919.x.

Thromboelastography in Patients with Acute Ischemic Stroke

Andrea Elliott, MD¹, Jeremy Wetzel, BA¹, Tiffany Roper, BA¹, Evan Pivalizza, MBChB¹, James McCarthy, MD¹, Cristina Wallace, MD¹, Mary Jane Hess, BSN¹, Hui Peng, PhD², Mohammad H Rahbar, PhD², Navdeep Sangha, MD¹, and James C Grotta, MD¹

many acute ischemic stroke patients (20 of 49 patients in our cohort) are hypercoagulable on various TEG parameters



Clinical Trial

> [Anesth Analg.](#) 2005 Jun;100(6):1576-83.

doi: 10.1213/01.ANE.0000155290.86795.12.

Thromboelastography maximum amplitude predicts postoperative thrombotic complications including myocardial infarction

Douglas J McCrath ¹, Elisabetta Cerboni, Robert J Frumento, Andrew L Hirsh, Elliott Bennett-Guerrero



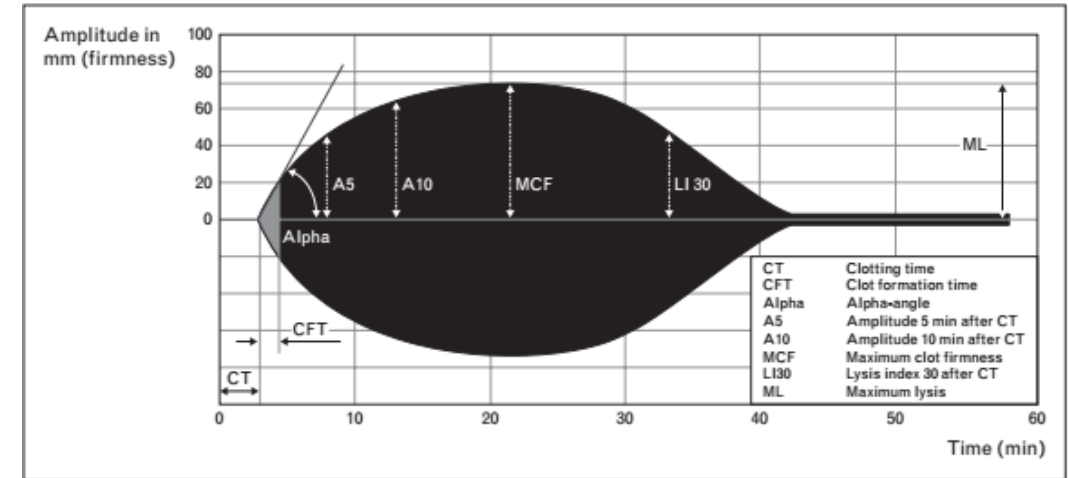
→ Extem (mimetiza "vida real")

→ Demora pra começar >> Ct largo >>> Fator de coagulação

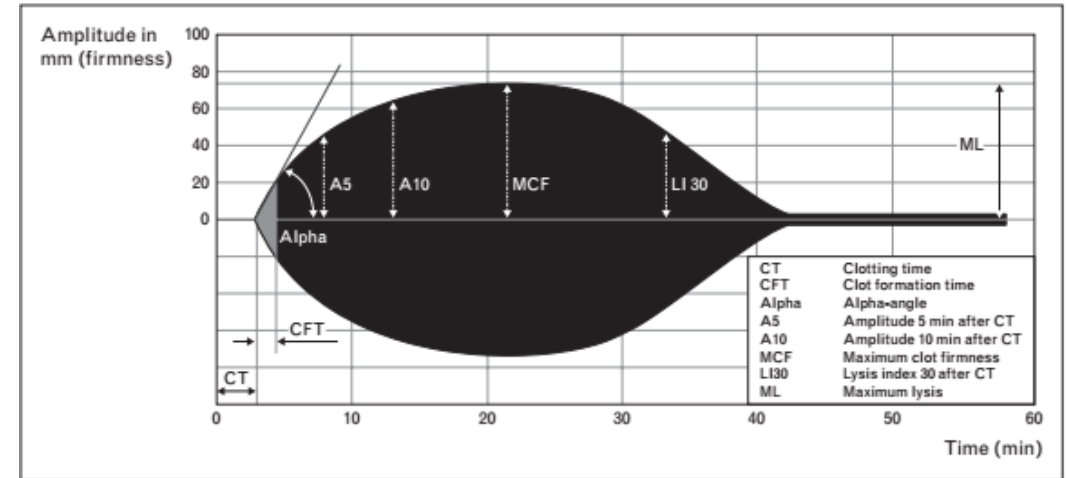
→ Trombo fino >> MCF baixo. >> Plaquetas ou fibrinogênio (FIBTEM)

→ Trombo ok >> Lise intensa >> Hiperfibrinólise (APTEM ?)

→ Uso de heparina (CEC?) >> INTEM alterado >> Heptem corrige



Ainda Mais Básico...



- Começa devagar >> Fator da coag.
- Não Progride > Fibrinogênio ou PLQ (olhar fibtem)
- Dissolve >> Hiperfibrinólise

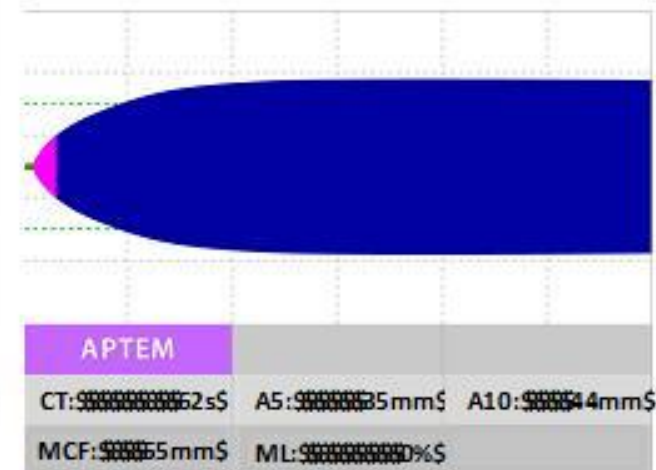
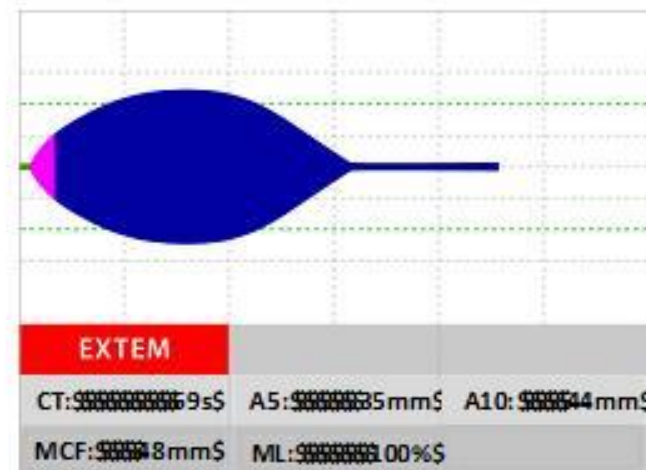
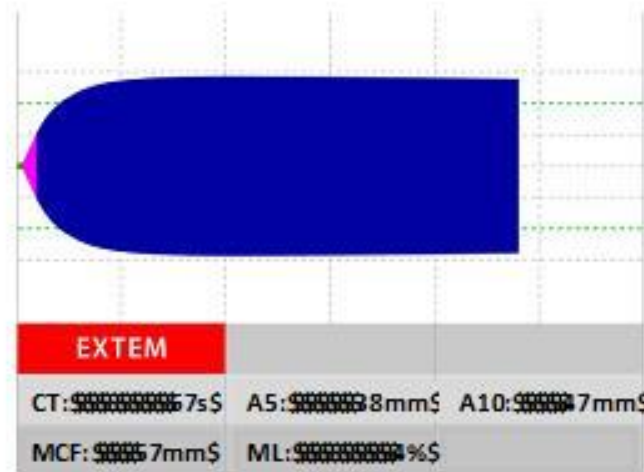


Exercícios!



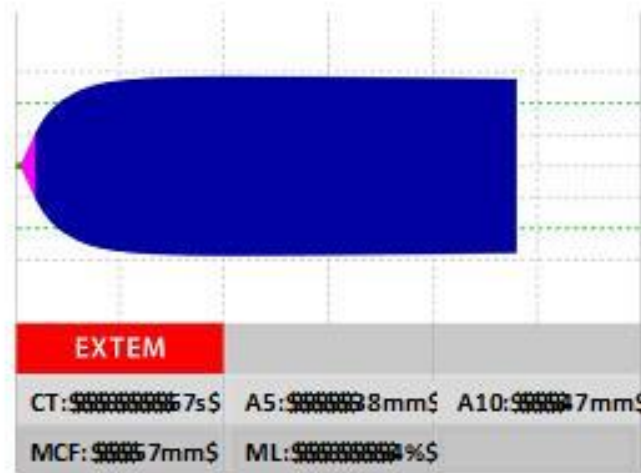
Normal clot:

EXTEM CT	43-82 s
EXTEM A10	40-60 mm
EXTEM MCF	52-70 mm
EXTEM ML	< 15%
EXTEM LI60	> 85%
FIBTEM A10	6-21 mm
FIBTEM MCF	7-24 mm



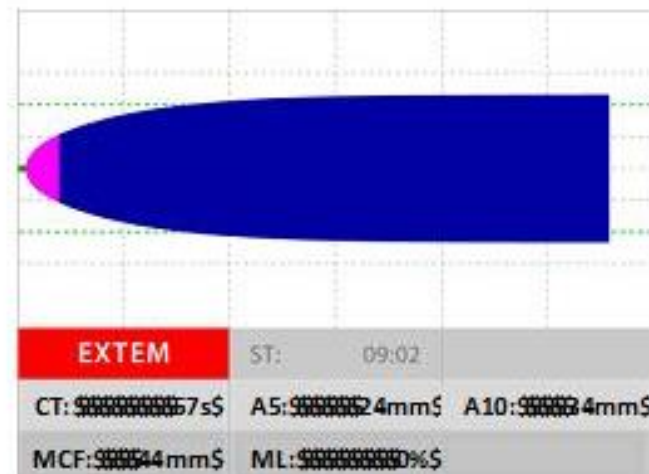
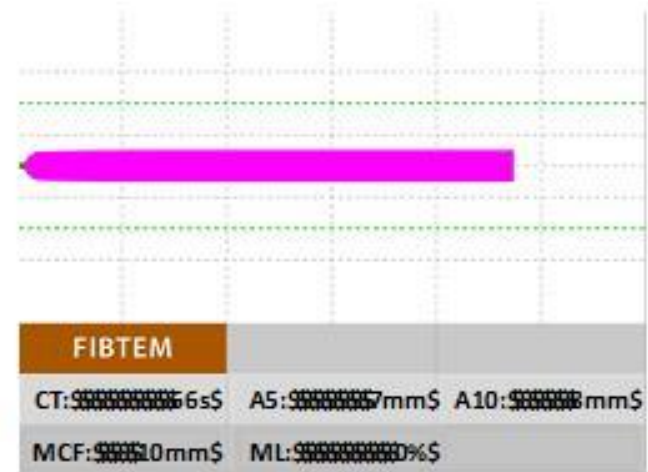
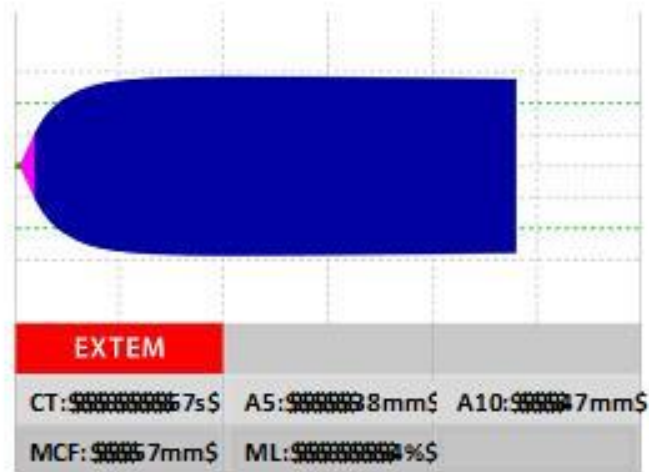
Normal clot:

EXTEM CT	43-82 s
EXTEM A10	40-60 mm
EXTEM MCF	52-70 mm
EXTEM ML	< 15%
EXTEM LI60	> 85%
FIBTEM A10	6-21 mm
FIBTEM MCF	7-24 mm



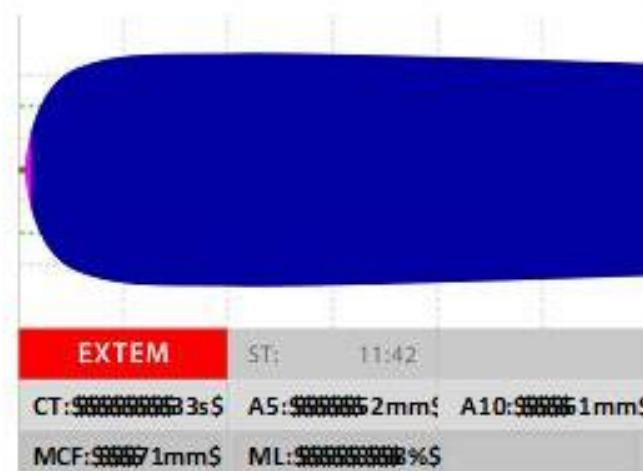
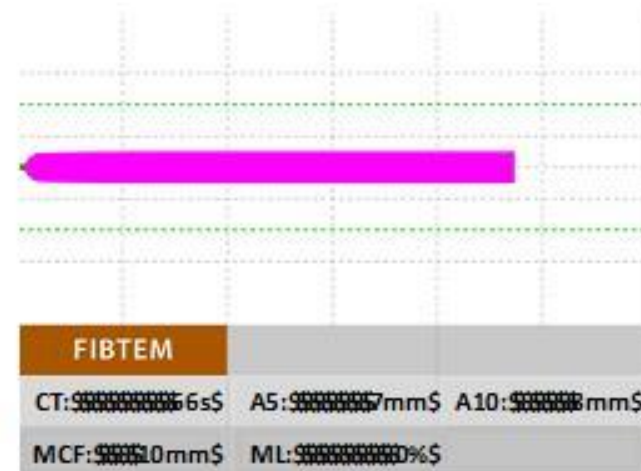
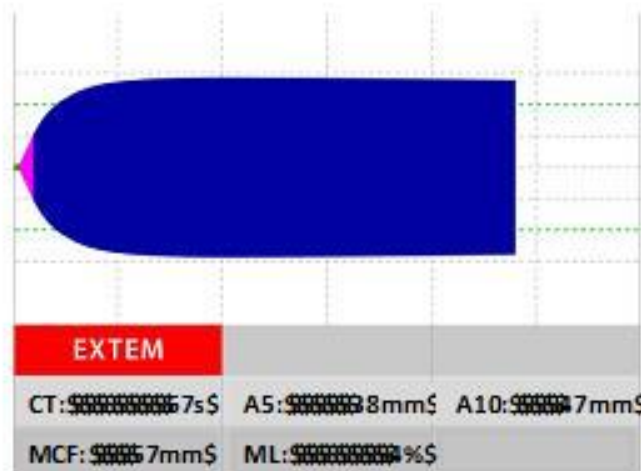
Normal clot:

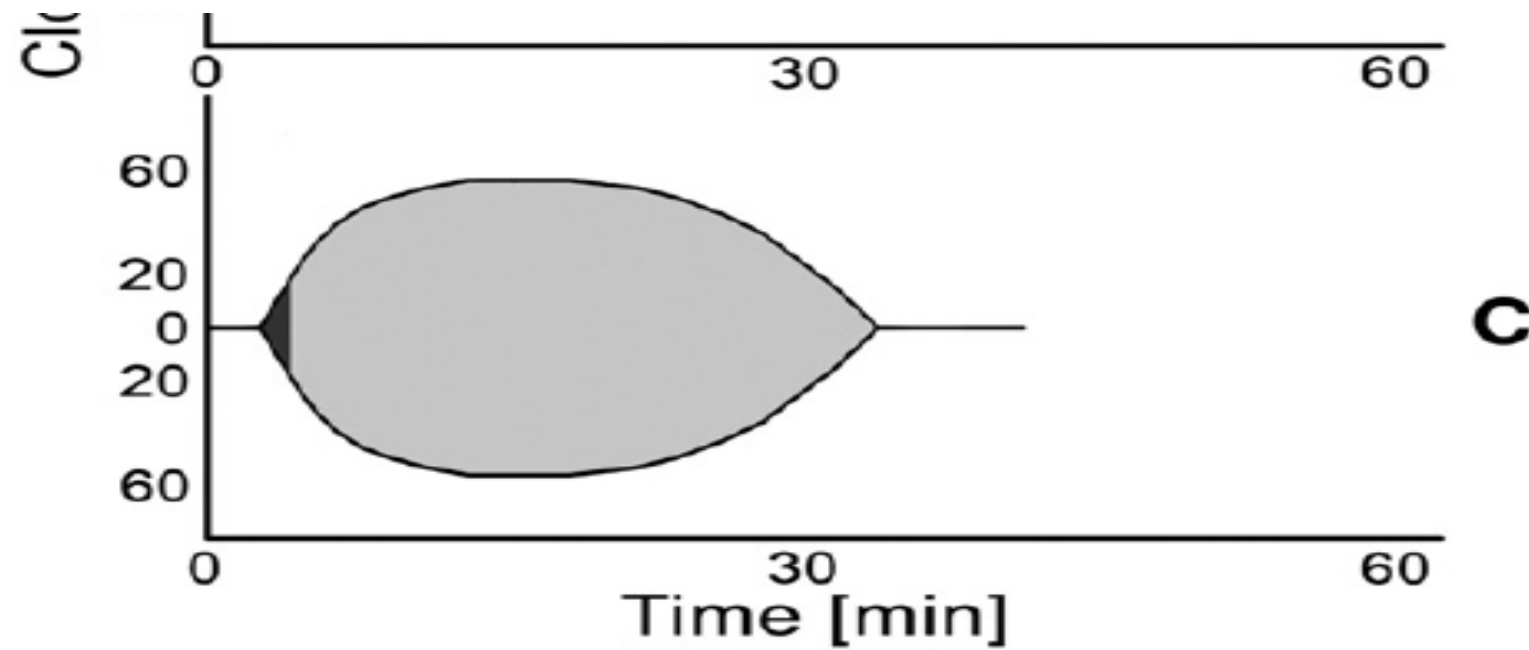
EXTEM CT	43-82 s
EXTEM A10	40-60 mm
EXTEM MCF	52-70 mm
EXTEM ML	< 15%
EXTEM LI60	> 85%
FIBTEM A10	6-21 mm
FIBTEM MCF	7-24 mm



Normal clot:

EXTEM CT	43-82 s
EXTEM A10	40-60 mm
EXTEM MCF	52-70 mm
EXTEM ML	< 15%
EXTEM LI60	> 85%
FIBTEM A10	6-21 mm
FIBTEM MCF	7-24 mm





hyperfibrinolysis (panel C).

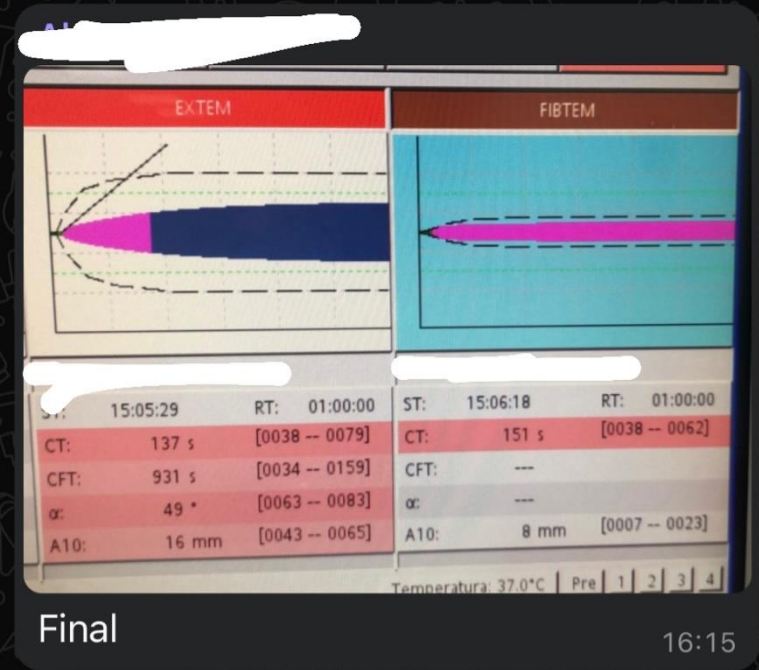


Pós Operatório de Transplante Hepático

Coagulopatia x Cirúrgico ??

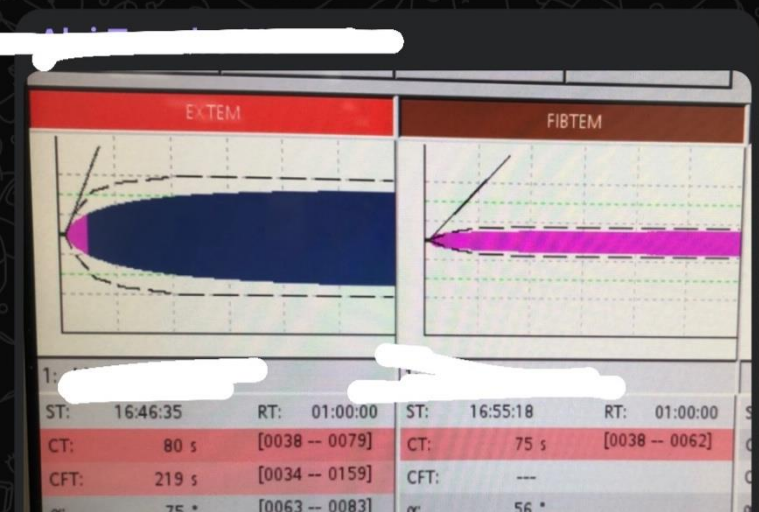
Parcial

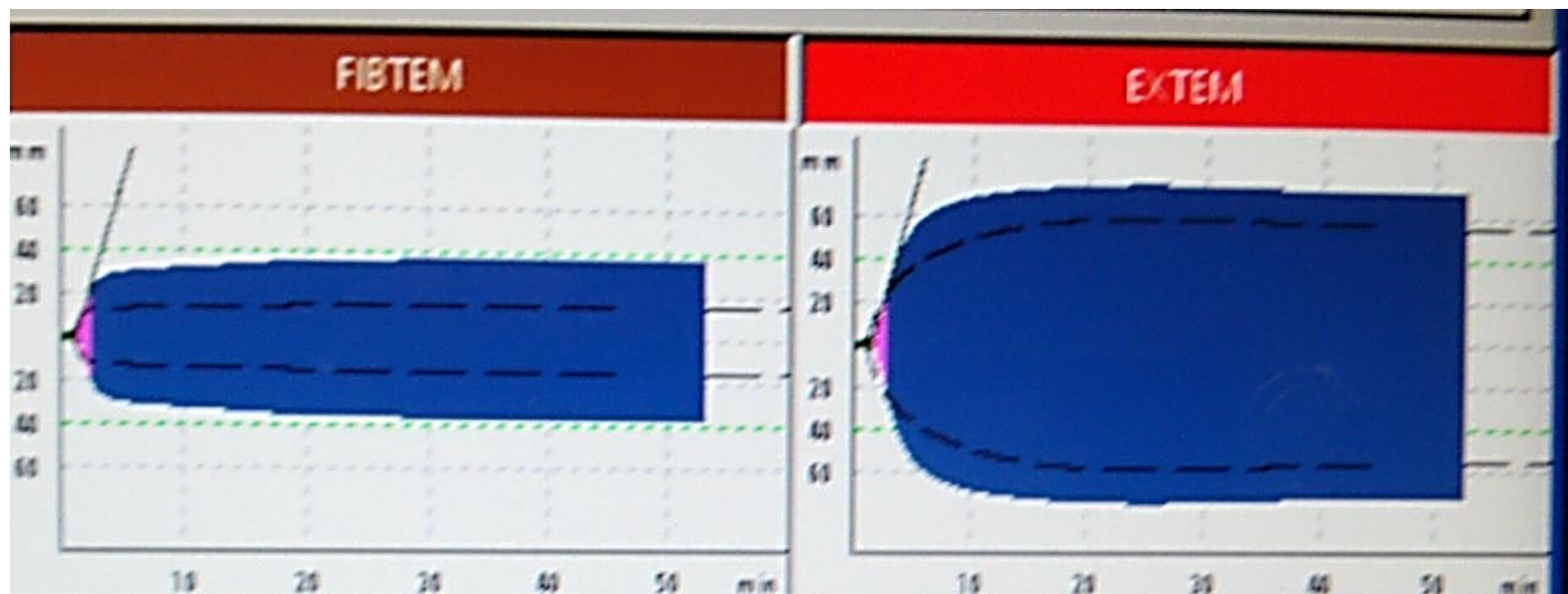
15:31



16:15

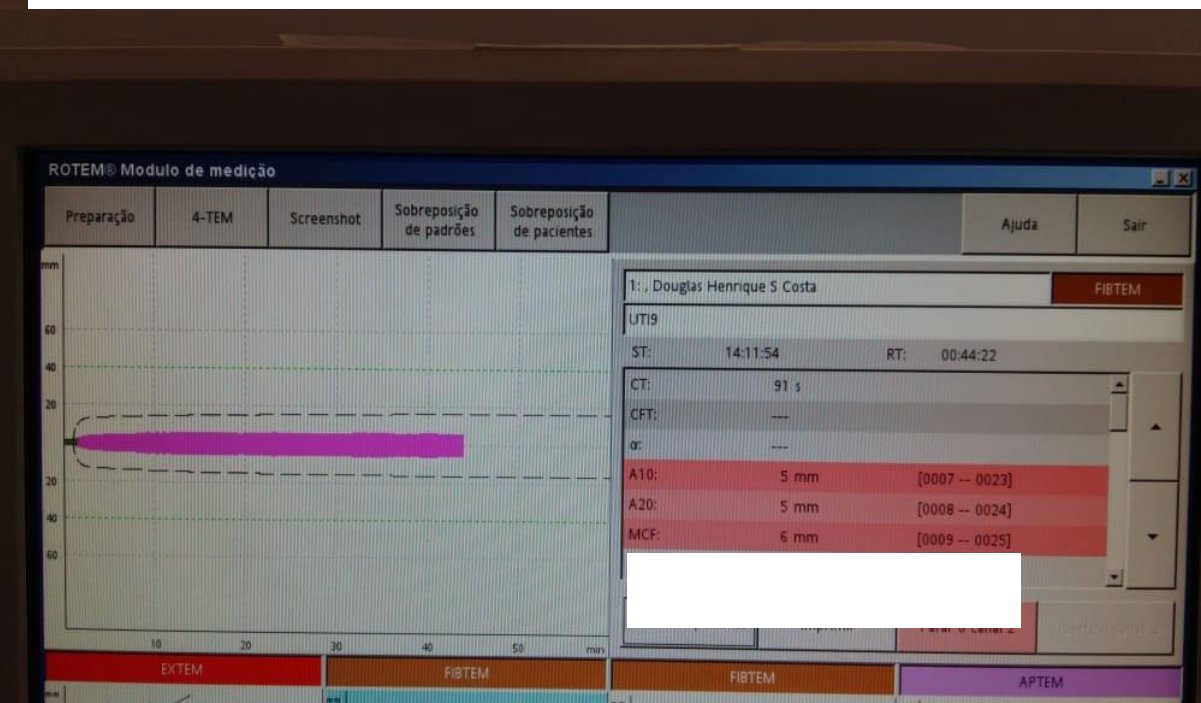
Mensagem não lida: 1



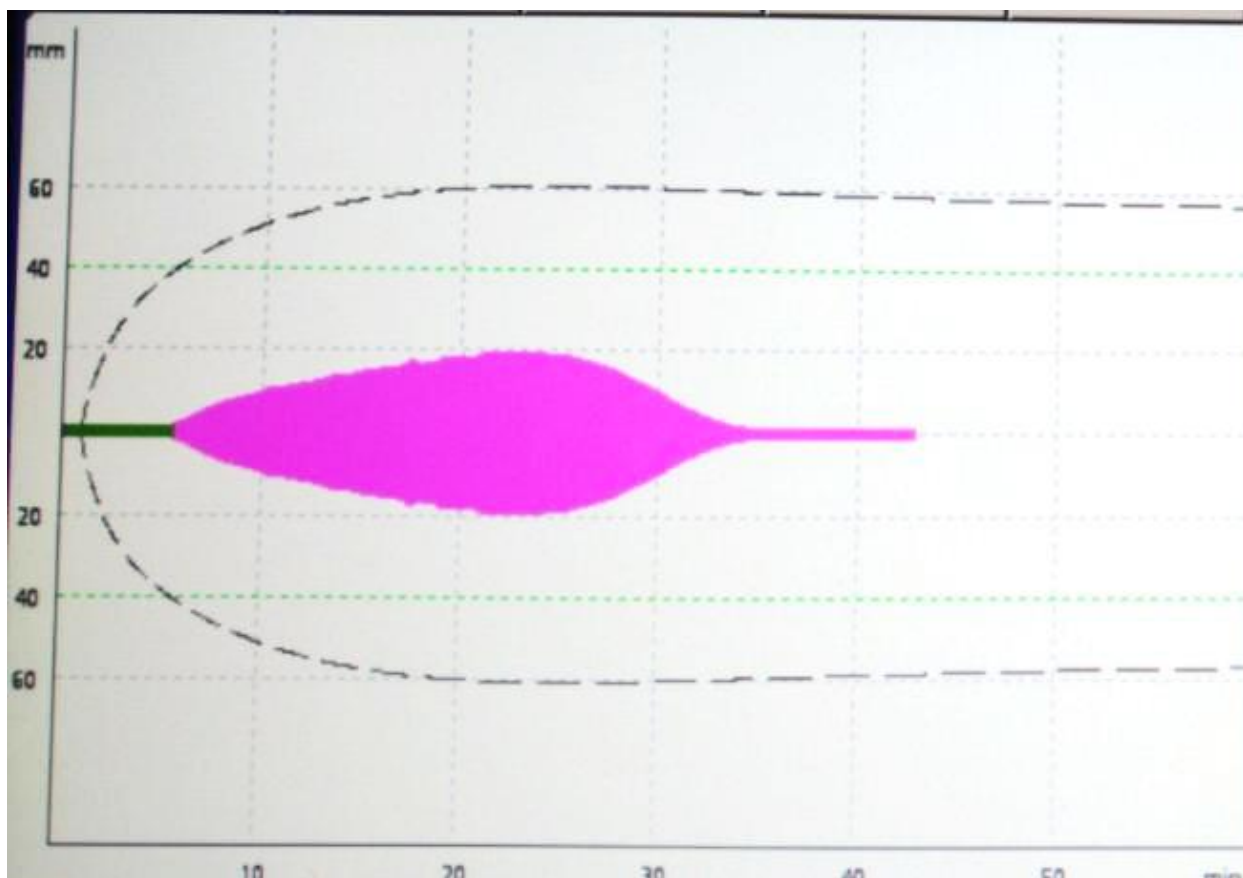


ST:	16:00:23	RT:	00:53:36	ST:	16:01:08	RT:	00:52:50
CT:	88 s			CT:	109 s	[0038 -- 0079]	
CFT:	59 s			CFT:	51 s	[0034 -- 0159]	
α :	79 °			α :	80 °	[0063 -- 0083]	
A10:	33 mm	[0007 -- 0023]		A10:	72 mm	[0043 -- 0065]	









1: aline dos santos lima,

EXTEM

UTI

43:18

CT: 338 s [0038 -- 0079]

CFT: ---

α : ---

A10: 15 mm [0043 -- 0065]

A20: 19 mm [0050 -- 0071]

MCF: 20 mm [0050 -- 0072]

ML: 100 %

Dados do paciente

Imprimir

Parar o canal 3

Libere o canal 3



ROTEM® Modulo de medição

Preparação

4-TEM

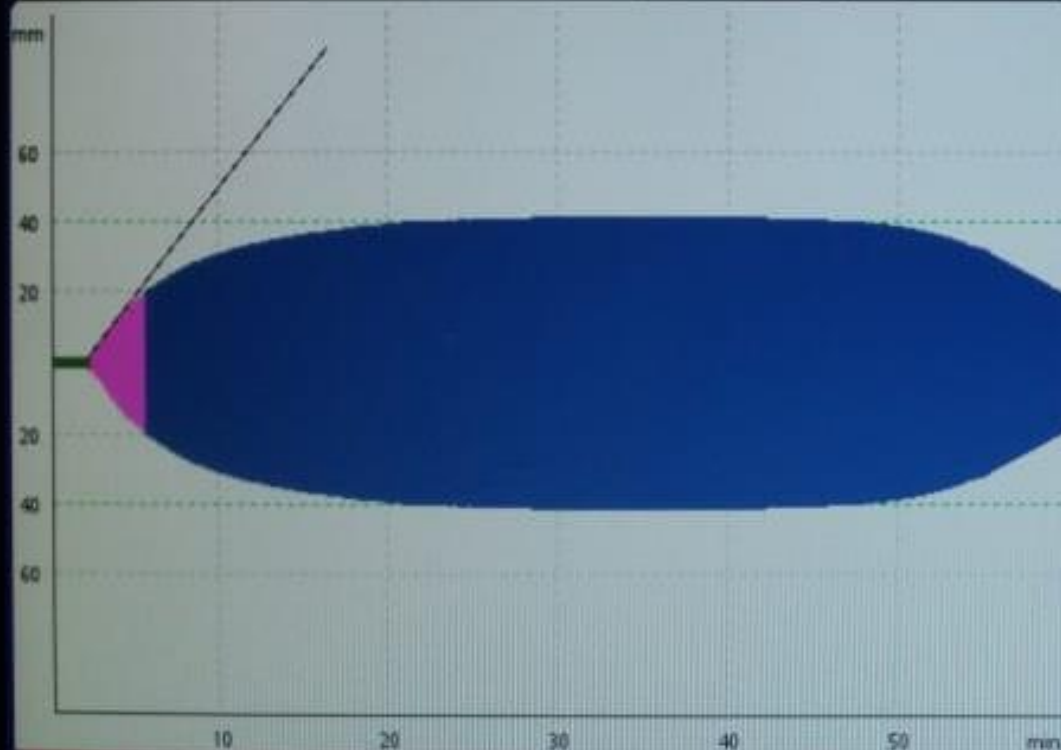
Screenshot

Sobreposição
de padrões

Sobreposição
de pacientes

Ajuda

Sair



T: Marisa B Freitas

EXTEM

UTI9

ST: 15:52:57

RT: 01:15:38

CFT: 200 s [0034 -- 0159]

α : 58 ° [0063 -- 0083]

A10: 34 mm [0043 -- 0065]

A20: 40 mm [0050 -- 0071]

MCF: 42 mm [0050 -- 0072]

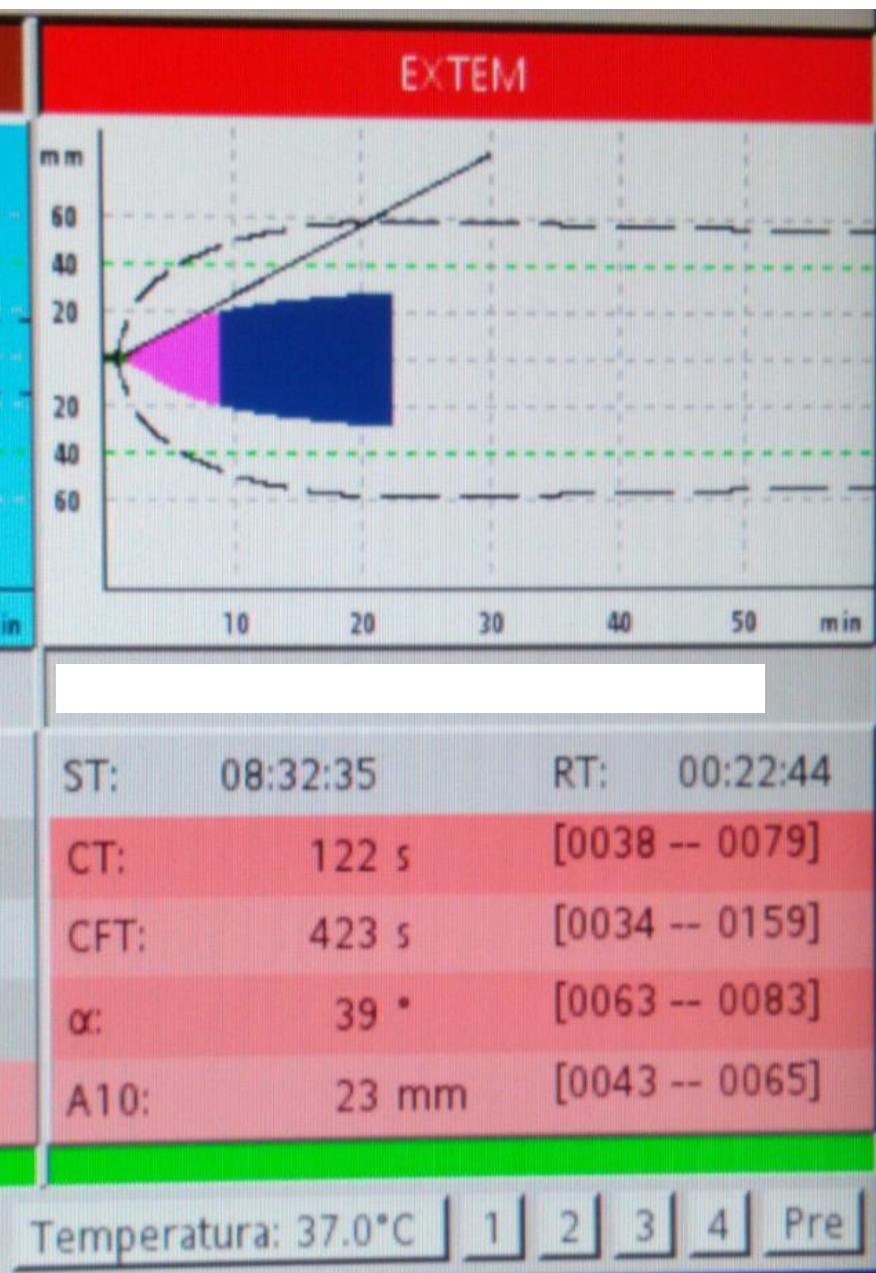
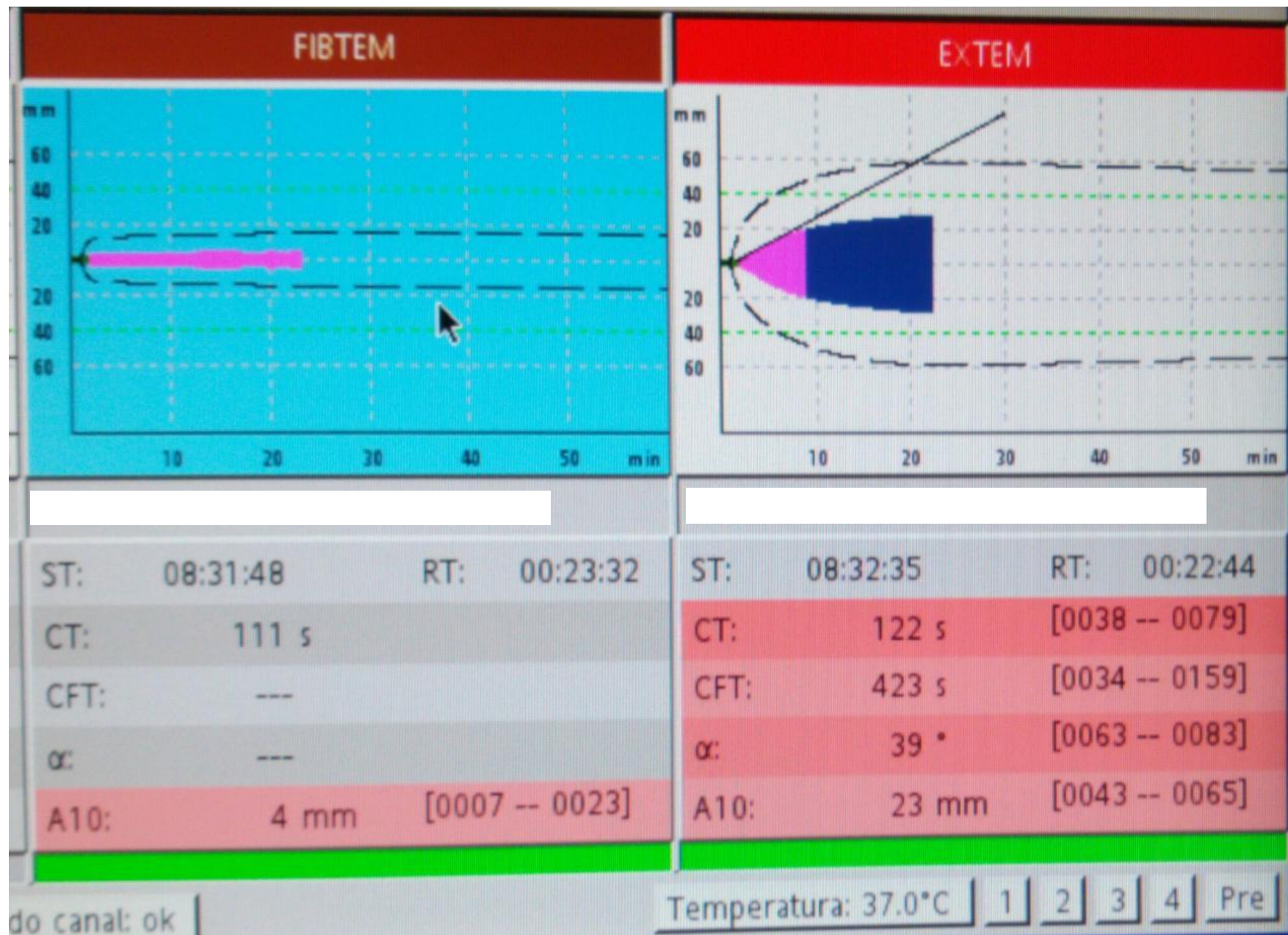
ML: * 99 %

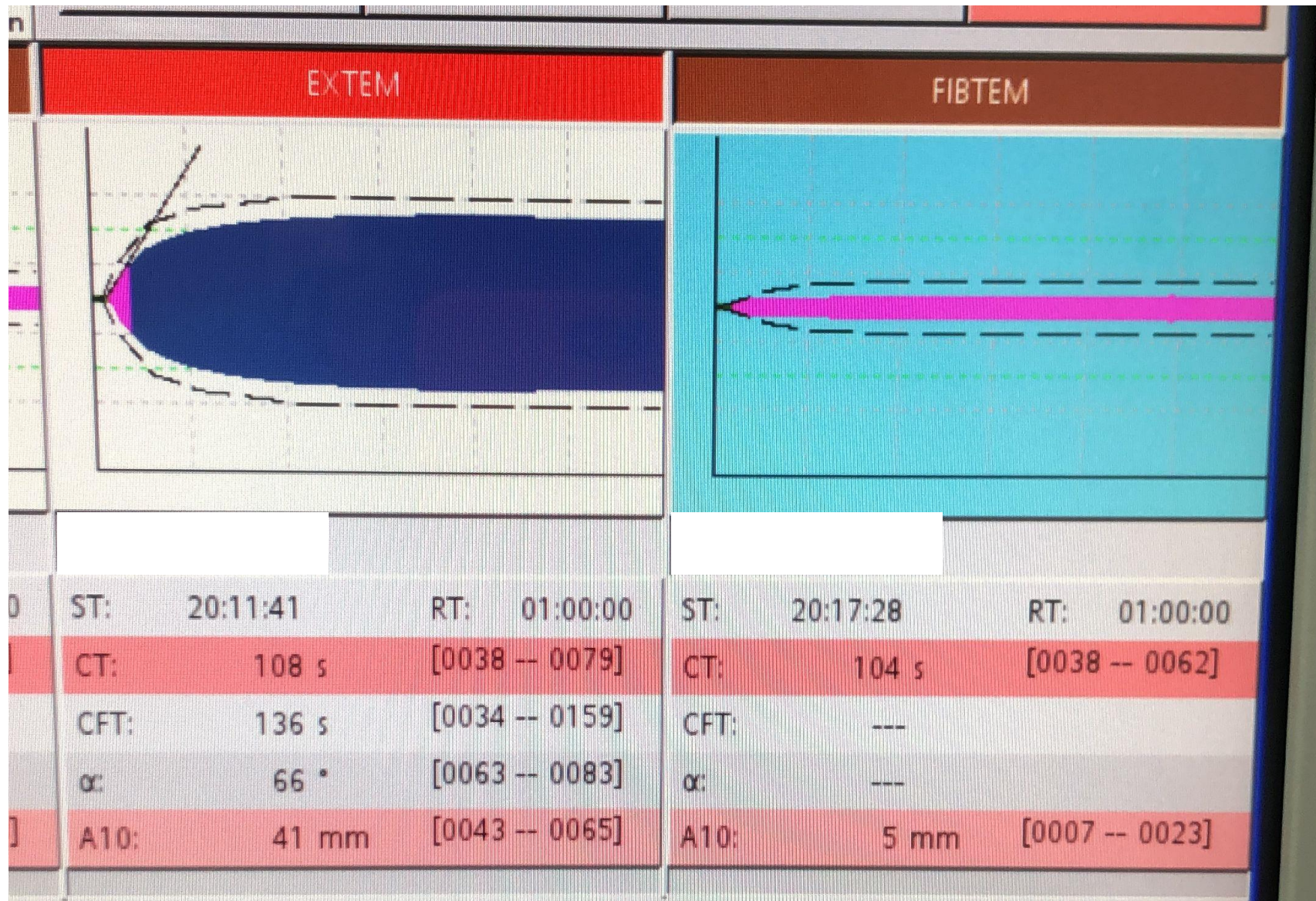
Dados do paciente

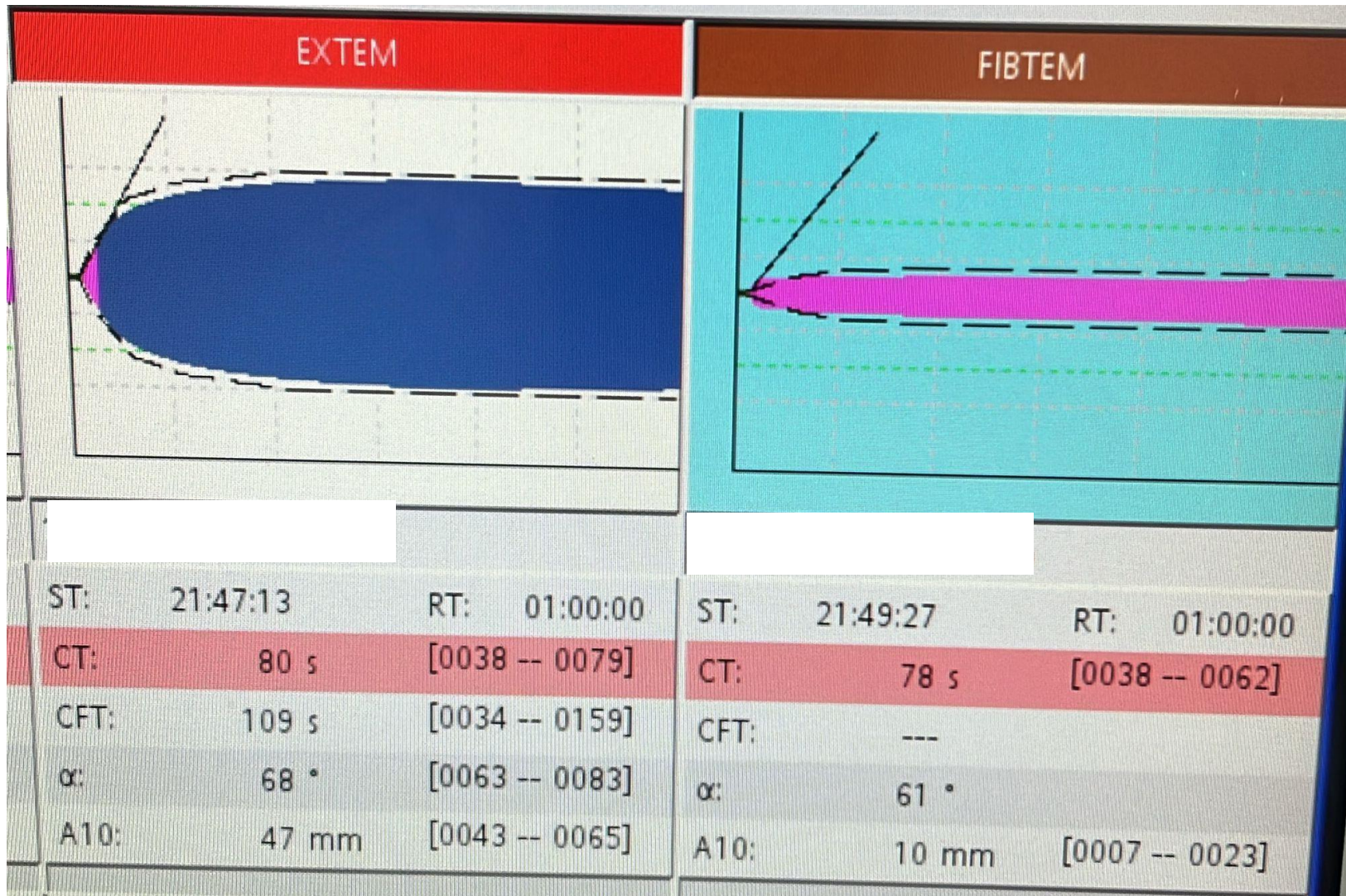
Imprimir

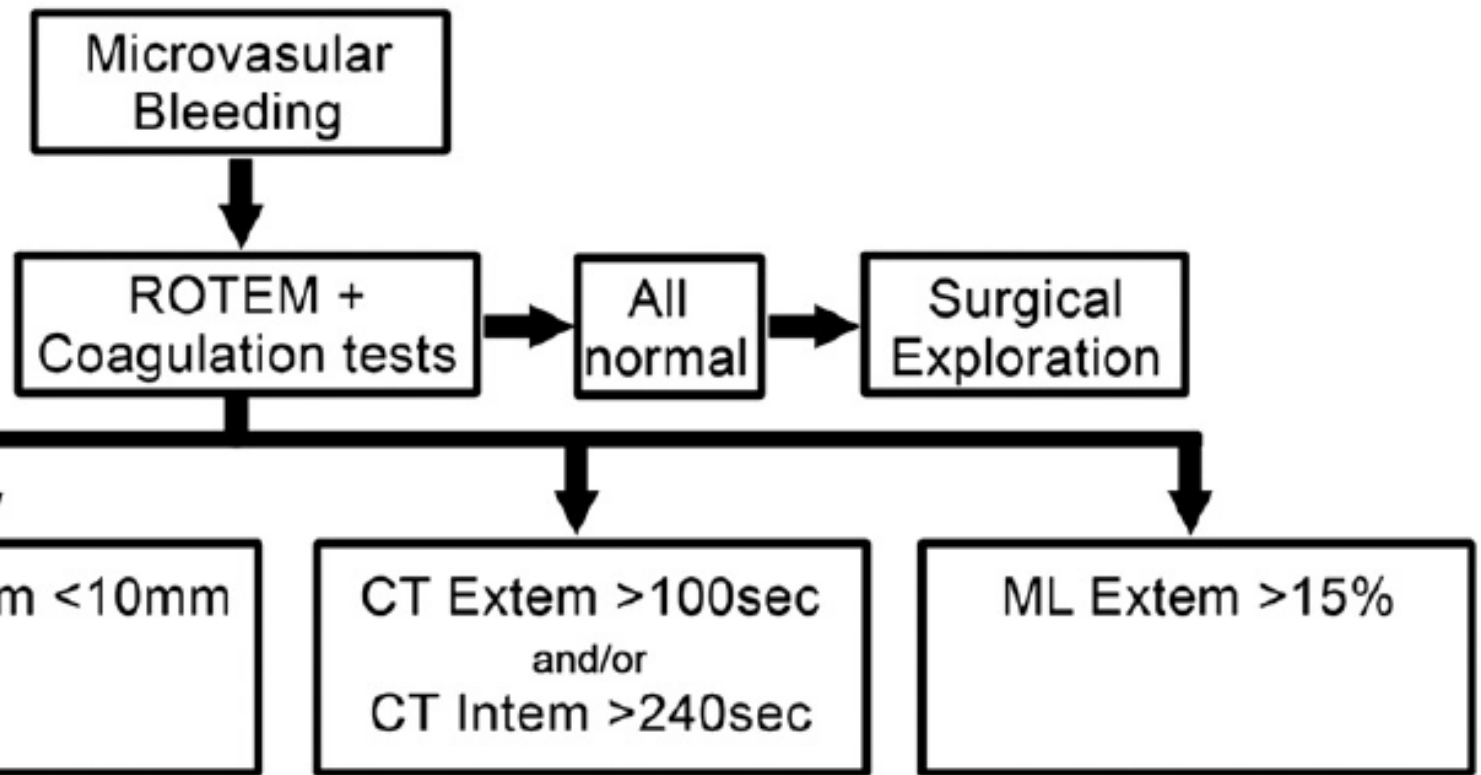
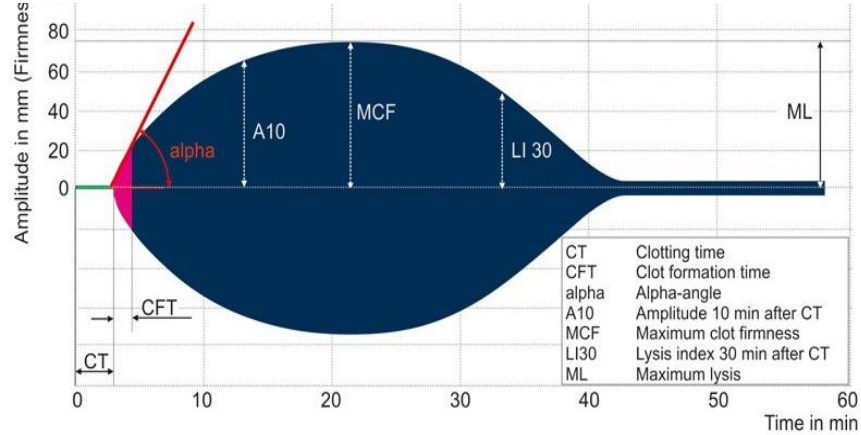
Parar o canal 3











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Obrigado !

